

FIT3152 Data Analytics - Assignment 2

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1. Data Exploration

The dataset includes 20,000 observations and 35 variables from the World Values Survey (WVS). Country and Wave variables identify each respondent's country and the survey wave. The other 33 variables include 30 predictors and 3 binary class variables.

The predictor variables cover several themes: political engagement and attitudes (PolArmy, PolBoycott, PolExperts, PolLeader, PolDemons, PolDemoc, ILPolitics), work and economic values (ILWork, HardWork, FutureWork, Competition, PrivateState, GovtRespons), leisure and social values (ILLeisure, ILFriends, ILReligion), civic participation (ACTArtsEd, ACTSport, ACTProfessional, ACTPolParty, ACTHumanitarian), personality traits (ICQTolerance, ICQDetermination, ICQHardWork, ICQIndependence, ICQThrift, ICQObedience, ICQFaith), and general well-being (LifeSatis, Age). The three class variables, CWomensMvt, CGovernment, and CUnions, each show whether respondents have high (1) or low (0) confidence in a social institution.

Table 1. Class variable proportions (High = 1, Low = 0)

Class Variable	Number of High	Number of Low	Proportions High	Proportions Low	Imbalance Ratio
CWomensMvt	9160	6418	0.5880	0.4120	1.427
CGovernment	8067	9267	0.4654	0.5346	1.149
CUnions	7162	10349	0.4090	0.5910	1.445

Table 1 shows the distribution of High and Low confidence ratings for the three class variables. CWomensMvt has the highest response rate (58.8%), CGovernment is relatively balanced (46.5% High), and CUnions is the most imbalanced, with 40.9% High responses. However, all imbalance ratios are below 1.5, showing only mild class imbalance.

Table 2. Summary statistics of 30 predictor attributes (N=20000)

Variable	Mean	SD	Min	Median	Max	N Missing	N Negative
PolArmy	1.862	2.727	-5	3	4	0	4074
ILPolitics	2.301	1.702	-5	3	4	0	1217
ILWork	1.230	1.394	-5	1	4	0	1123
HardWork	2.521	4.224	-5	3	10	0	4426
ICQTolerance	0.643	0.580	-5	1	1	0	114
ILReligion	1.661	1.635	-5	2	4	0	1170
LifeSatis	6.657	2.616	-5	7	10	0	285
ICQDetermination	0.237	0.861	-5	0	1	0	582

PolBoycott	1.933	1.851	-5	3	3	0	2338
ICQHardWork	0.487	0.689	-5	1	1	0	235
ICQIndependence	0.435	0.596	-5	0	1	0	132
PolExperts	1.330	2.275	-5	2	4	0	4021
ICQThrift	0.335	0.577	-5	0	1	0	137
ACTArtsEd	-0.718	1.877	-5	0	2	0	4833
FutureWork	2.018	1.398	-5	3	3	0	1095
ILLeisure	1.613	1.454	-5	2	4	0	1064
PolLeader	1.677	2.402	-5	2	4	0	3648
Competition	2.775	3.491	-5	3	10	0	2829
ACTSport	-0.732	1.993	-5	0	2	0	5136
ACTProfessional	-0.762	1.837	-5	0	2	0	4716
PolDemons	1.831	1.780	-5	2	3	0	2084
ACTPolParty	-0.760	1.807	-5	0	2	0	4620
PrivateState	4.552	3.783	-5	5	10	0	2351
GovtRespons	5.530	3.746	-5	6	10	0	1406
PolDemoc	0.858	1.973	-5	1	4	0	3510
ACTHumanitarian	-0.753	1.858	-5	0	2	0	4760
ICQObedience	0.340	0.579	-5	0	1	0	142
Age	41.268	16.891	-5	39	99	0	229
ILFriends	1.460	1.359	-5	2	4	0	928
ICQFaith	0.324	0.679	-5	0	1	0	268

Table 2 has no NA entries; however, all variables have minimum values of -5 due to WVS coding convention, where negative integers (-1 to -5) represent different types of missing responses (-1 = Don't know, -2 = No answer/refused, -3 = Not applicable, and -5 = Missing / not applicable for other reasons).

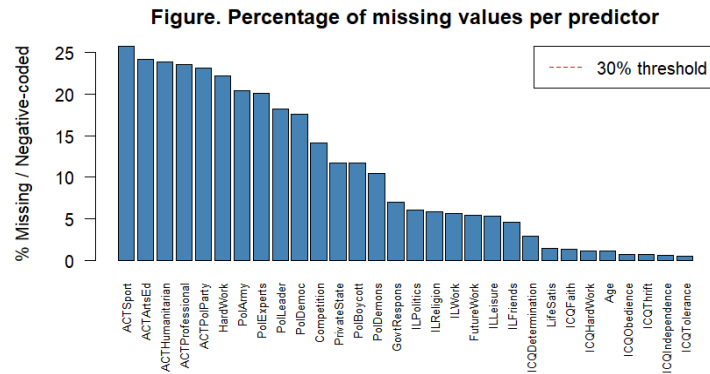


Figure 1. Percentage of missing values per predictor

In Figure 1, the civic participation variables ACTSport (25.7%), ACTArtsEd (24.2%), ACTHumanitarian (23.8%), ACTProfessional (23.6%), and ACTPolParty (23.1%) exhibit the highest proportions of negative-coded values. Similarly, political attitude variables PolArmy (20.4%), HardWork (22.1%), and PolExperts (20.1%) show elevated missingness. ICQ personality traits and Age have low missingness (under 2%) and show high response completion. Despite this, all variables remain below 30% threshold, and none were removed.

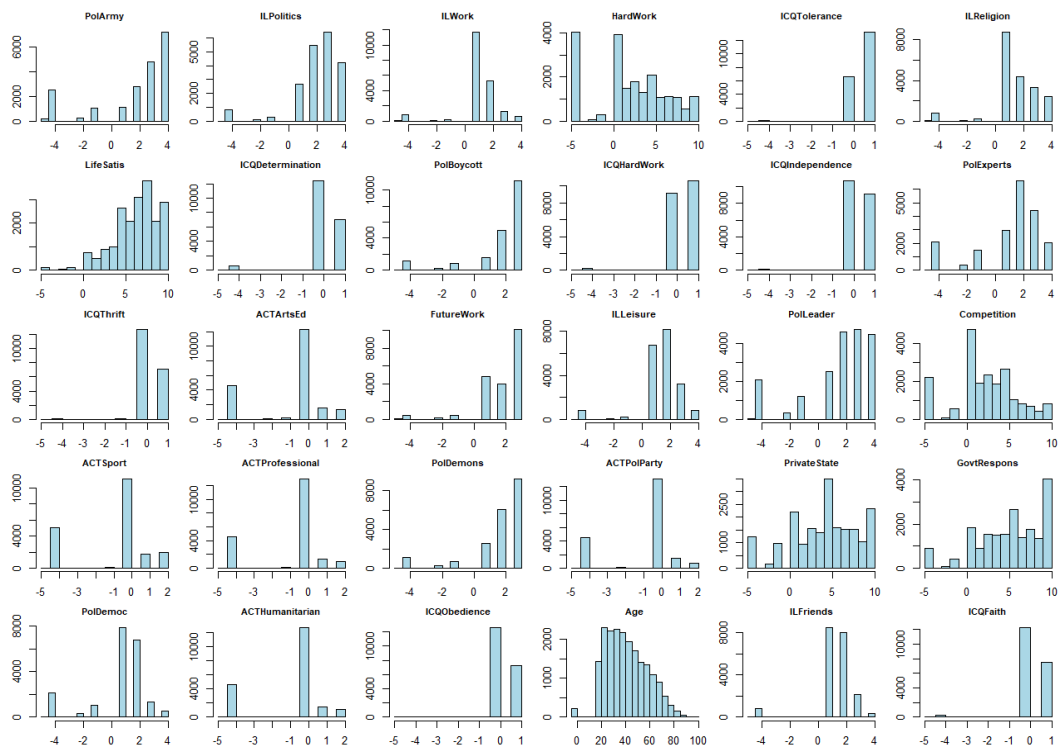


Figure 2: Histogram for all numeric predictors

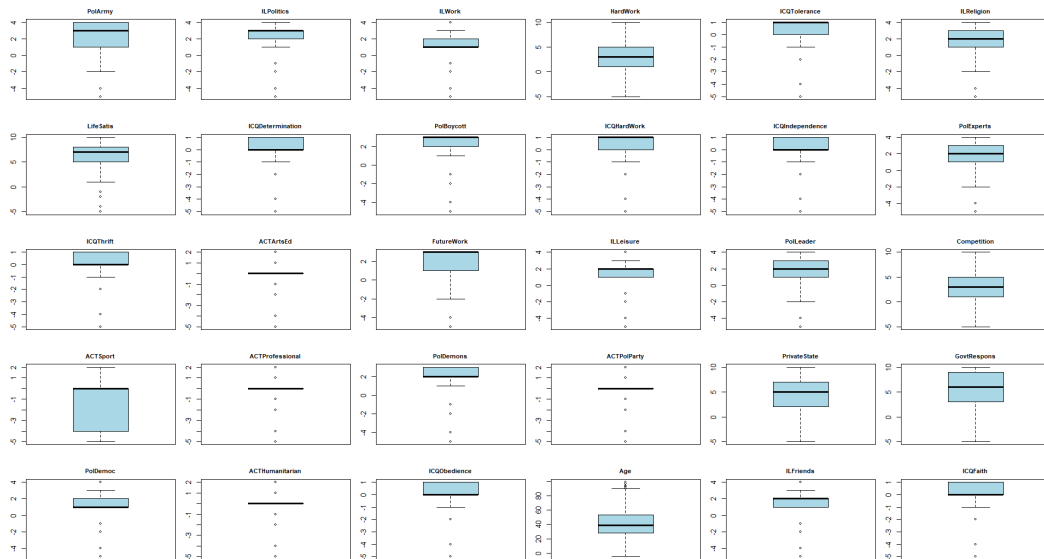


Figure 3: Box plot for all numeric predictors

Figures 2 and 3 together show that most ordinal predictors are left-skewed or multimodal, typical for Likert-scale survey data, with extremely low outliers confirming the presence of WVS negative codes to be replaced by NA. ACT membership variables show near-constant distributions after imputation, showing limited discriminatory power. In contrast, Age (right-skewed, no missing values), PrivateState, GovtRespons, and LifeSatis show variation with wide interquartile ranges, consistent with their high feature importance rankings in Q8.

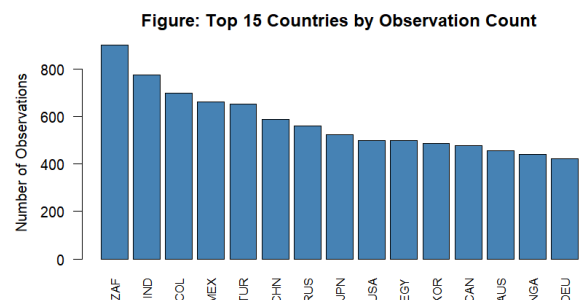


Figure 4: Top 15 Countries by Observation Count

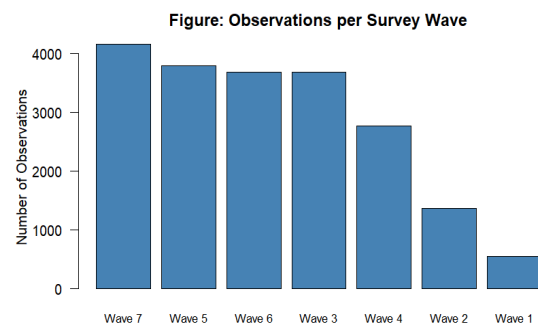


Figure 5: Observations per Survey Wave

The dataset includes 58 countries across 7 survey waves. In Figure 4, South Africa (ZAF) has the highest number of observations, followed by India (IND) and Colombia (CLD), while Finland and Slovakia have the fewest (Appendix: Figure ??). In Figure 5, Wave 7 contains the most observations, while Wave 1 has the fewest. Not all countries appear in every wave, resulting in an unbalanced longitudinal dataset.

2. Pre-processing

Pre-processing was then carried out in several steps. Negative WVS codes were replaced with NA, and rows with missing class labels were removed via listwise deletion using `complete.cases()`. Missing predictor values were then imputed using median values, as it is more robust to skewed distributions and outliers. Finally, class variables were converted to **binary factors** (“Low” and “High”), and Country and Wave were converted to categorical factors to reflect their identifier nature.

Table 3. Dataset summary after pre-processing

Item	Value
Total observations	14,112
Predictor columns	30
Class variables converted	CWomensMct, CGovernment, CUnions
Remaining NAs	0

3. Train Test Splitting

The dataset was divided into training and test sets using a 70/30 split, with a fixed random seed (`set.seed(33986010)`) to ensure reproducibility. This produced 9,878 training observations and 4,234 test observations. Predictor variables were selected by excluding Country, Wave, and the three class variables to avoid data leakage and ensure only relevant features were used for model training.

4. Classification Model Implementation

Five classifiers were trained on the 70% training set for each of the three binary targets (CWomensMvt, CGovernment, CUnions), with class variables excluded from predictors to prevent data leakage.

a. Decision Tree

Using the *rpart* package with 10-fold cross-validation pruning, **CWomensMvt** and **CUnions** collapsed to a single root node with no splits as no predictor attribute provided sufficient Gini reduction to improve out-of-sample generalisation.

Figure: Decision Tree - CGovernment

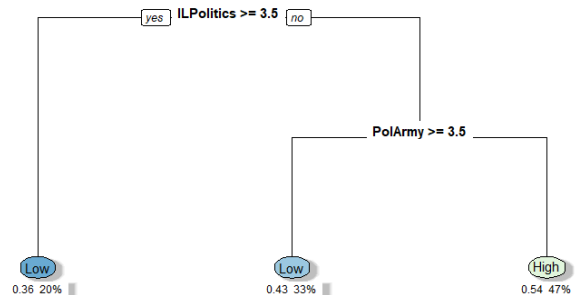


Figure 6: Decision Tree - CGovernment

For CGovernment, the pruned tree produced three terminal nodes (Figure 6) splitting on **ILPolitics** and **PolArmy**. Respondents who rated politics as important ($ILPolitics \geq 3.5$, 20% of data) tend toward Low government confidence (64% Low, 36% High), suggesting political engagement is associated with institutional scepticism. Among the remaining 80%, stronger support for army rule ($PolArmy \geq 3.5$) predicts High confidence, reflecting a positive relationship between authoritarian attitudes and government trust.

b. Naïve Bayes

Fitted using `e1071::naiveBayes()` on all 30 predictors. Assumes conditional independence between predictors and serves as a probabilistic baseline classifier.

c. Bagging (Bootstrap Aggregating)

Implemented using `adabag::bagging()` with 25 bootstrap trees (`mfinal = 25`), aggregating predictions by majority vote to reduce variance.

d. Boosting (AdaBoost)

Applied using `adabag::boosting()` with 25 iterations. Trees are built sequentially, with up-weighting misclassified observations at each step to reduce bias.

e. Random Forest

Random Forest model was trained using `randomForest()` with 100 trees (`ntree = 100`) and `mtry = 5`.

Table 4: Random Forest OOB error estimates during training.

Class Variable	OOB Error Rate	Low Class Error	High Class Error
CWomensMvt	41.74%	69.48%	22.02%
CGovernment	42.42%	33.42%	52.79%
CUnions	41.81%	21.91%	69.40%

OOB error rates of about 41–43% show that the models have moderate difficulty predicting all target variables. CWomensMvt and CUnions have much higher errors for the minority classes (about 69–70%), showing majority-class bias. CGovernment has more balanced errors because its class distribution is more symmetric.

5. Classification & Performance Metrics

The performance of all five classifiers was evaluated on the 30% test set.

CGovernment

Table 5: Performance Metrics

Model	Accuracy	Precision	Recall	F1
Decision Tree	0.5650	0.5233	0.5519	0.5372
Naive Bayes	0.5864	0.5534	0.4972	0.5238
Bagging	0.5834	0.5697	0.3650	0.4449
Boosting	0.5964	0.5743	0.4548	0.5076
Random Forest	0.5983	0.5703	0.4941	0.5295

Table 6: Confusion Matrices for all models

	Actual Class									
	Decision Tree		Naive Bayes		Bagging		Boosting		Random Forest	
Predicted	Low	High	Low	High	Low	High	Low	High	Low	High
Low	1323	868	1520	974	1763	1230	1644	1056	1576	980
High	974	1069	777	963	534	707	653	881	721	957

CGovernment is the most balanced class variable (imbalance ratio = 1.15), resulting in more reliable model performance. The Decision Tree achieved the highest recall (0.552) and a

competitive F1-score (0.537), while Random Forest obtained the highest accuracy (0.599). Naïve Bayes showed the best recall among ensemble models, whereas Bagging performed the weakest. Overall, CGovernment produced the most consistent results because of its class distribution.

CUnions

Table 7: Performance Metrics

Model	Accuracy	Precision	Recall	F1
Decision Tree	0.5987	0.0000	0.0000	0.0000
Naive Bayes	0.5888	0.4798	0.2931	0.3639
Bagging	0.6162	0.5640	0.1919	0.2863
Boosting	0.6205	0.5561	0.2684	0.3620
Random Forest	0.6190	0.5499	0.2790	0.3702

Table 8: Confusion Matrices for all models

	Actual Class									
	Decision Tree		Naive Bayes		Bagging		Boosting		Random Forest	
	Predicted Low	Predicted High	Predicted Low	Predicted High	Predicted Low	Predicted High	Predicted Low	Predicted High	Predicted Low	Predicted High
Low	2535	1699	1995	1201	2283	1373	2171	1243	2147	1225
High	0	0	540	498	252	326	364	456	388	474

The Decision Tree collapsed into a single-node model that predicted only the majority Low class, resulting in zero precision, recall, and F1 for the High class. This suggests that the model was biased toward the majority class despite the dataset having only mild class imbalance identified in Q1. Among the other models, Random Forest achieved the highest F1 (0.371), followed closely by Naïve Bayes and Boosting. However, all models showed low recall (~0.19–0.29), indicating poor detection of high-confidence cases. Although Boosting achieved the highest accuracy (0.621), this was due to correct predictions of the majority of the low class.

CWomensMvt

Table 9: Performance Metrics

Model	Accuracy	Precision	Recall	F1
Decision Tree	0.5905	0.5905	1.0000	0.7425
Naive Bayes	0.5739	0.6400	0.6364	0.6382
Bagging	0.5900	0.5958	0.9500	0.7323
Boosting	0.6108	0.6205	0.8772	0.7269
Random Forest	0.6006	0.6181	0.8468	0.7146

Table 10: Confusion Matrices for all models

	Actual Class									
	Decision Tree		Naive Bayes		Bagging		Boosting		Random Forest	
Predicted	Low	High	Low	High	Low	High	Low	High	Low	High
Low	0	0	839	909	123	125	393	307	426	383
High	1734	2500	895	1591	1611	2375	1341	2193	1308	2117

The Decision Tree predicted every observation as High, having perfect recall (1.000), but poor classification performance due to its single-node tree. Therefore, its high F1-score (0.7425) is misleading. Naïve Bayes showed the most balanced performance (F1 = 0.638). Bagging, Boosting, and Random Forest achieved higher F1-scores (around 0.72–0.73) with very high recall but low precision, showing the models tended to predict many observations as High. Overall, all models had difficulty correctly identifying low-confidence cases for CWomensMvt.

Overall Summary

Overall, CGovernment gave the most balanced and reliable results because its classes are nearly even. CUnions was the hardest target to predict because of consistently low recall for the High class. Although CWomensMvt achieved the highest F1 scores, these were largely inflated by majority-class bias. Decision Trees performed poorly due to collapsing into single-node models for CWomensMvt and CUnions. Random Forest showed the most consistent overall performance across all targets, while Naïve Bayes gave the best balance between precision and recall, particularly for imbalanced classes.

6. ROC Curves and AUC Analysis

ROC curves and AUC values were used to evaluate the discriminative performance of the five classifiers across the three class variables. An AUC of 1.0 indicates perfect classification, while 0.5 represents random guessing.

Table 11: AUC

(a) CWomensMvt		(b) CGovernment		(c) CUnions	
Model	AUC	Model	AUC	Model	AUC
Decision Tree	0.5000	Decision Tree	0.5754	Decision Tree	0.5000
Naive Bayes	0.5740	Naive Bayes	0.6111	Naive Bayes	0.5847
Bagging	0.5694	Bagging	0.6067	Bagging	0.5953
Boosting	0.6026	Boosting	0.6267	Boosting	0.6084
Random Forest	0.5940	Random Forest	0.6259	Random Forest	0.6088

CGovernment

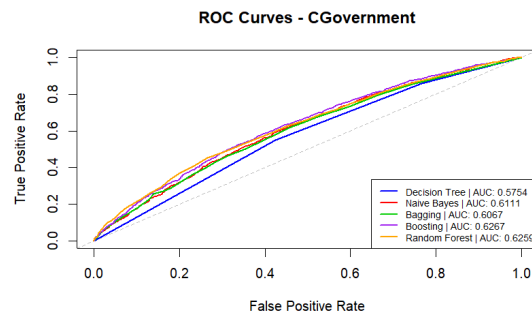


Figure 7: ROC Curves - CGovernment

All models achieved their strongest performance, reflecting the more balanced class distribution. Boosting (0.627) and Random Forest (0.626) produced the highest AUC values, while the Decision Tree performed lower (0.575) but still above random chance.

CUnions

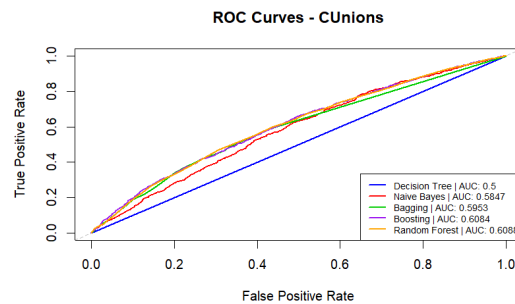


Figure 8: ROC Curves - CUnions

The Decision Tree again produced AUC = 0.500, while Random Forest (0.609) and Boosting (0.608) achieved the best performance. Overall, CUnions was more difficult to classify than CGovernment but slightly easier than CWomensMvt.

CWomensMvt

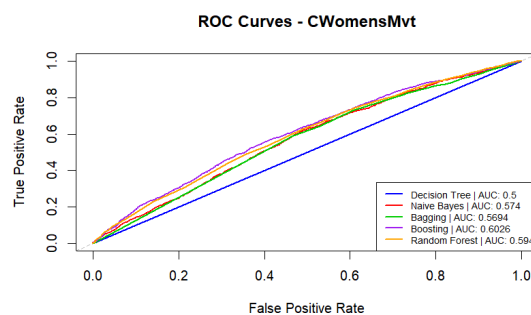


Figure 9: ROC Curves - CWomensMvt

the Decision Tree achieved AUC = 0.500 due to its single-node structure, indicating random performance. Boosting achieved the highest AUC (0.603), followed by Random Forest (0.594), while all models showed only moderate discrimination due to weak predictor signal and mild class imbalance.

Overall AUC Summary

Overall, Boosting achieved the highest average AUC (0.613), closely followed by Random Forest (0.610). Since all AUC values were between 0.50 and 0.63, the predictors have only limited ability to distinguish between classes.

7. Comparison of Classifier Performance

Table 12: Full Metrics - CGovernment

	Accuracy	Precision	Recall	F1	AUC
Decision Tree	0.5650	0.5233	0.5519	0.5372	0.5754
Naive Bayes	0.5864	0.5534	0.4972	0.5238	0.6111
Bagging	0.5834	0.5697	0.3650	0.4449	0.6067
Boosting	0.5964	0.5743	0.4548	0.5076	0.6267
Random Forest	0.5983	0.5703	0.4941	0.5295	0.6259

Table 13: Full Metrics - CUnions

	Accuracy	Precision	Recall	F1	AUC
Decision Tree	0.5987	0.0000	0.0000	0.0000	0.5000
Naive Bayes	0.5888	0.4798	0.2931	0.3639	0.5847
Bagging	0.6162	0.5640	0.1919	0.2863	0.5953
Boosting	0.6205	0.5561	0.2684	0.3620	0.6084
Random Forest	0.6190	0.5499	0.2790	0.3702	0.6088

Table 14: Full Metrics - CWomensMvt

	Accuracy	Precision	Recall	F1	AUC
Decision Tree	0.5905	0.5905	1.0000	0.7425	0.5000
Naive Bayes	0.5739	0.6400	0.6364	0.6382	0.5740
Bagging	0.5900	0.5958	0.9500	0.7323	0.5694
Boosting	0.6108	0.6205	0.8772	0.7269	0.6026
Random Forest	0.6006	0.6181	0.8468	0.7146	0.5940

Table 15: Best and Worst Classifiers by F1

ClassVar	Best	BestF1	Worst	WorstF1
CWomensMvt	Decision Tree	0.7425	Naive Bayes	0.6382
CGovernment	Decision Tree	0.5372	Bagging	0.4449

CUnions	Random Forest	0.3702	Decision Tree	0.0000
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Table 16: Average F1 per Ensemble Model

	Ensemble	Avg_F1
Bagging	Bagging	0.4878
Boosting	Boosting	0.5322
Random Forest	Random Forest	0.5381

Best classifier per class variable

CWomensMvt: The Decision Tree has the highest F1 (0.743), but it is misleading because it uses a single-node structure. The best model is Boosting (F1 = 0.727, AUC = 0.603)

CGovernment: Random Forest performs best overall (Accuracy = 0.598, F1 = 0.530, AUC = 0.626), while Boosting has the highest AUC (0.627)

CUnion: Random Forest is the best performer (F1 = 0.370, AUC = 0.609)

Overall, Boosting and Random Forest are the most reliable models across all three targets.

Worst Classifier & Ensemble Classifier

The **Decision Tree** is the weakest model overall. It fails for CWomensMvt and CUnions (AUC = 0.500) and also has the lowest AUC for CGovernment (0.575).

Among ensemble methods, **Bagging performs the worst**, with the lowest average F1 (0.488) and weak recall across all class variables. This happens because it keeps class imbalance in each bootstrap sample, leading to biased predictions toward the majority class.

Do Best/Worst Classifiers Vary Across Class Variables?

Yes, performance changes across class variables due to differences in class balance.

Table 17: Best & Worst Classifiers Across Class Variables

Class Variable	Best (meaningful)	Worst
CWomensMvt	Boosting (AUC = 0.603)	Decision Tree (AUC = 0.500)
CGovernment	Boosting/Random Forest (AUC ≈ 0.626–0.627)	Decision Tree (AUC = 0.575)
CUnions	Random Forest (AUC = 0.609)	Decision Tree (AUC = 0.500)

Overall, **Boosting and Random Forest** are consistently the best models, while the **Decision Tree** is consistently the worst. **Bagging** is the weakest ensemble method overall, especially struggling with low recall for the minority High class. Performance is highest for CGovernment because it is the most balanced target variable, while CUnions and CWomensMvt, being more imbalanced, produce lower and less reliable metrics.

Investigative Tasks

8. Feature Importance

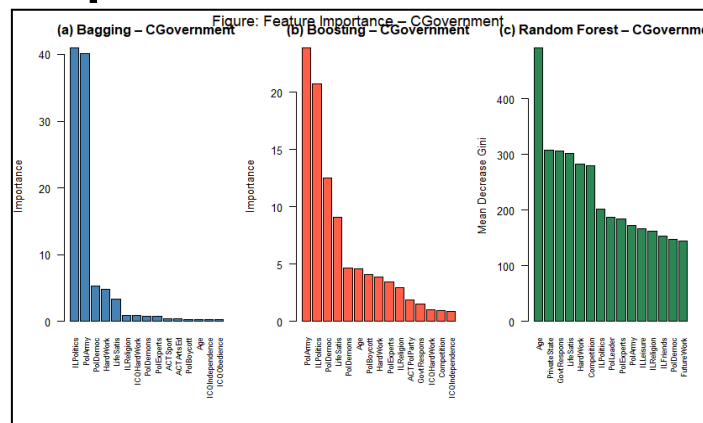


Figure 10: Feature Importance - CGovernment

Most important attributes: ILPolitics and PolArmy are the top predictors in Bagging and Boosting, showing that political views strongly influence confidence in government. PrivateState (private vs state ownership of business) is also important across models, suggesting economic ideology (state vs market) matters. Age is again the strongest predictor in Random Forest, while LifeSatis and GovtRespons also appear consistently.

Can be removed: ICQIndependence, ICQObedience, ACTPolParty, and ACTHumanitarian have very low importance across all ensemble models and can be removed with minimal performance effect

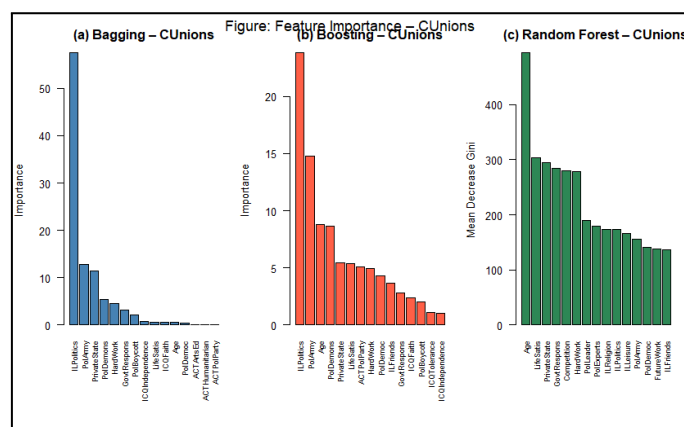


Figure 11: Feature Importance - CUnions

Most important attributes: PolArmy, PolDemoc, and ILPolitics are consistently top across all ensemble models, showing that political attitudes are the main drivers of confidence in government. Age is also important in Random Forest, suggesting generational differences matter. GovtRespons and LifeSatis also contribute, showing general satisfaction and economic views.

Can be removed: ICQIndependence, ICQObedience, ACTSport, and ACTArtsEd have very low importance across models and could be removed with minimal impact on performance.

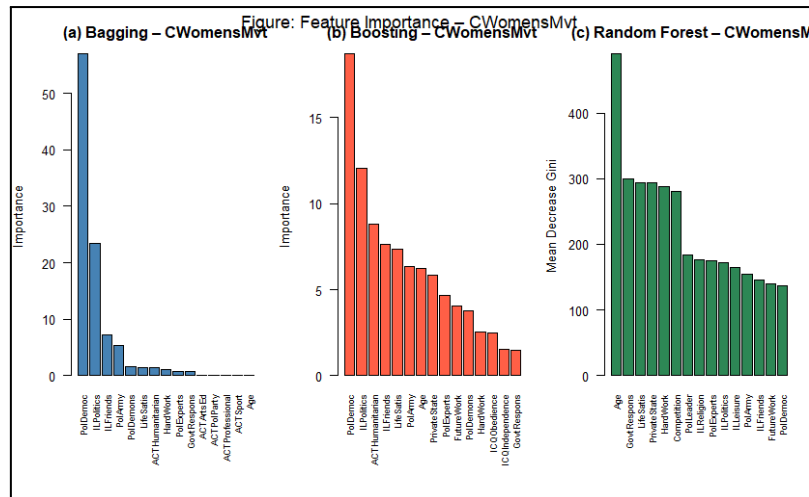


Figure 12: Feature Importance - CWomensMvt

Most important attributes: PolDemoc and ILPolitics are the top predictors in Bagging and Boosting, showing that political views are the main factor in confidence in the Women’s Movement. Age is the strongest predictor in Random Forest, suggesting generational differences matter. LifeSatis is also consistently important across all models.

Can be removed: ACTPolParty, ACTProfessional, ICQObedience, and Age (in Bagging/Boosting) have very low importance and contribute little to the models, so they could be removed with minimal effect on performance.

Summary

Table 18: Key Predictor Variables and Their Roles Across Classification Models

Attribute	Role	Consistent across
Age	Dominant in Random Forest for all 3	All 3 class variables
ILPolitics	Top predictor in Bagging & Boosting	All 3 class variables
PolArmy	Strong predictor of authority-related confidence	All 3 class variables
PolDemoc	Democratic attitudes	CWomensMvt, CGovernment
LifeSatis	General wellbeing	All 3 class variables
GovtRespons	Economic/governance views	CGovernment, CUnions
PrivateState	Economic ideology	CUnions

Conversely, child quality attributes (**ICQIndependence**, **ICQObedience**, **ICQHardWork**) and organisational membership variables (**ACTSport**, **ACTArtsEd**, **ACTProfessional**) show very low importance across all models and class variables. This suggests they contribute little to prediction and could be removed with minimal impact.

9.Explaining Differences in Model Performance

Decision Tree uses simple feature splits and cannot capture complex relationships between predictors (Breiman et al., 1984). In this WVS dataset, political attitude variables interact

non-linearly, so a single tree cannot capture these relationships well. It performs poorly on CWomensMvt and CUnions, where it collapses into a single node (AUC = 0.500). It only performs better on CGovernment due to its more balanced class distribution (F1 = 0.537), showing that its performance depends heavily on the data structure.

Naïve Bayes assumes predictors are independent (Rish, 2001), which is not true for WVS data where political variables are correlated. Despite this, Naïve Bayes demonstrated the most balanced precision-recall trade-off across all three class variables. This is likely because it estimates class probabilities directly instead of drawing a hard decision boundary, making it less affected by the mild imbalance found in Q1. The AUC values (0.574–0.611) are modest but still show it picks up some signal, even with the violated assumption.

Bagging reduces variance by averaging bootstrap trees (Breiman, 1996), but it keeps class imbalance in each sample, leading to low recall in Q5 (CUnions: 0.192, CGovernment: 0.365). Correlated predictors also reduce tree diversity, shown in Q8, limiting its effectiveness (Galar et al., 2012). This explains why Bagging consistently underperforms Boosting and Random Forest.

Boosting improves performance by focusing on misclassified cases (Freund & Schapire, 1997). It handles non-linear relationships well and achieves the highest AUC across targets (0.603, 0.627, 0.608), and the reweighting also helps with class imbalance to some extent by giving minority-class errors more attention. However, it is sensitive to noise and outliers, mislabelled observations receive increasing weight over iterations, which might be why its edge over Random Forest is not that large on survey data.

Random Forest improves on Bagging by randomly limiting which features each tree can split on ($m_{try} = 5$), reducing correlation between trees (Breiman, 2001). The result is a more diverse ensemble that generalises better, which matches the consistently strong results in Q5 & Q6 (e.g. highest F1 for CUnions: 0.370 and CGovernment: 0.530). With 100 trees, individual noisy predictions tend to get cancelled out, making it more stable than either the Decision Tree or Boosting.

Why Performance Varies Across Class Variables

CGovernment (ratio 1.15) is more balanced, so all models learn better decision boundaries and achieve higher AUCs, capturing non-linear relationships more effectively.

CWomensMvt and **CUnions** (ratio ~1.44) have mild imbalance, which leads to poorer recall and even causes single trees to collapse. Models rely more on the majority class, weakening performance.

Overall, performance differences are explained by how well each model handles non-linearity, interactions, noise, and class imbalance.

10. Improving the Worst Ensemble Model

The worst-performing ensemble model was **Bagging**, as indicated by its lowest average F1-score across all three class variables in Question 7.

Improvement Strategies

Feature Selection: Instead of using 30 predictor attributes, it is reduced to 16 attributes (Age, LifeSatis, GovtRespons, PrivateState, HardWork, Competition, PolLeader, ILPolitics, PolExperts, ILReligion, ILLeisure, PolArmy, ILFriends, FutureWork, ILWork, PolDemoc) by taking the union of the top 15 most important predictors (by Mean Decrease Gini) from the Random Forest models fitted in Question 8, across all three class variables. This reduces noise and helps individual trees form cleaner decision boundaries.

Parameter Tuning: The number of base learners (**mfinal**) was increased from 25 to 100 to improve model stability and produce more reliable ensemble predictions.

Confusion Matrices

Table 19. Confusion Matrices — Improved Bagging Model

		CWomens Mvt		CGovern ment		CUnions	
		Actual Low	Actual High	Actual Low	Actual High	Actual Low	Actual High
Predicted	Low	115	112	1,624	1,095	2,330	1,454
Predicted	High	1,619	2,388	673	842	205	245

CWomensMvt

Figure: Q10 Improved Bagging – CWomensMvt

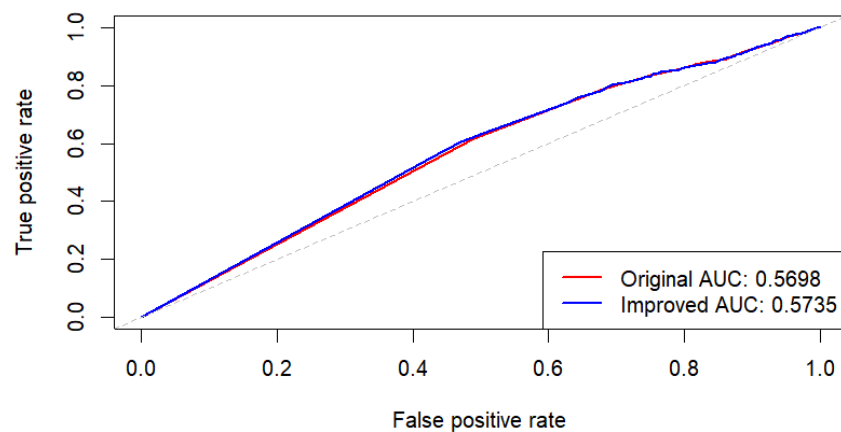


Figure 13: Improved Bagging - CWomensMvt

Table 20: Original vs Improved - CWomensMvt

Model	Accuracy	Precision	Recall	F1	AUC
Original	0.5869	0.5944	0.9456	0.730	0.5698
Improved	0.5912	0.5960	0.9552	0.734	0.5735

Improvement was modest (F1: 0.7300 → 0.7340), showing that Bagging was already performing well on this class variable and that the original feature set was largely sufficient.

CGovernment

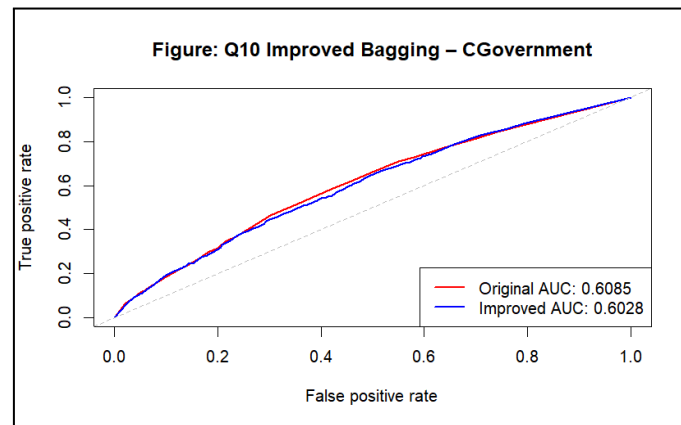


Figure 14: Improved Bagging CGovernment

Table 21: Original vs Improved - CGovernment

Model	Accuracy	Precision	Recall	F1	AUC
Original	0.5848	0.5770	0.3464	0.4329	0.6085
Improved	0.5824	0.5558	0.4347	0.4878	0.6028

The AUC slightly decreased (0.6085 → 0.6028) even though the F1 score improved. This indicates a trade-off, where the model shifted more predictions toward the “High” class, improving recall and F1 but slightly weakening its ability to rank classes effectively.

CUnions

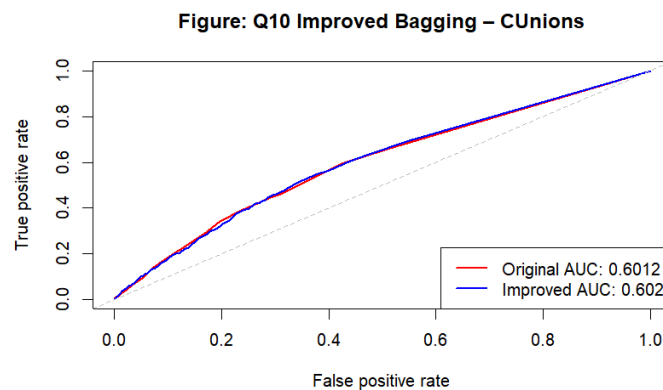


Figure 15: Improved Bagging - CUnions

Table 22: Original vs Improved - CUnions

Model	Accuracy	Precision	Recall	F1	AUC
Original	0.6117	0.5659	0.1389	0.2231	0.6012
Improved	0.6082	0.5444	0.1442	0.2280	0.6020

The improvement was minimal (F1: 0.2231 → 0.2280), with AUC staying almost the same (0.6012 → 0.6020). This aligns with earlier results showing that CUnions is the most

imbalanced and hardest class to predict, indicating that the limitation is mainly due to the data rather than the model setup.

Overall, feature selection and parameter tuning improved F1 scores across all three class variables. This shows that removing low-importance features and increasing the number of trees helps improve Bagging performance on this dataset.

11. Artificial Neural Network (ANN)

Country Selection

South Africa (ZAF) had the highest number of observations (548), as shown in Figure 16. It was followed by TUR (520), COL (486), USA (461), and MEX (447). **South Africa** was selected for ANN training and evaluation.

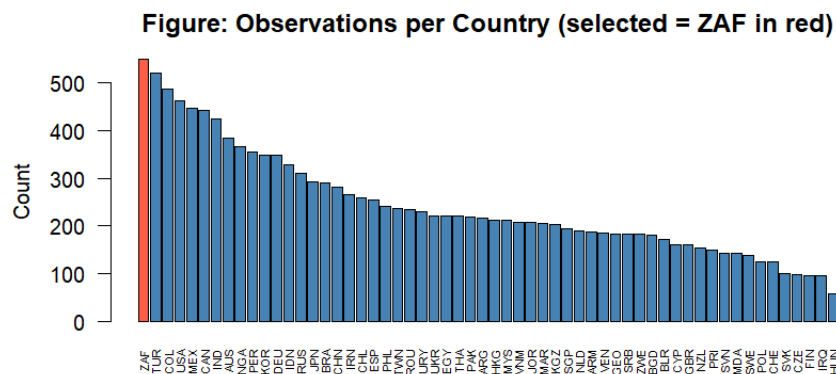


Figure 16: Observations per Country

The class variable selected for ANN classification was **CWomensMvt** (confidence in the Women's Movement), as it exhibited the highest overall predictability across models in Questions 5–6, providing the most meaningful baseline for ANN comparison.

Data Pre-processing

Predictor Selection: 11 key predictors were selected from the top-10 Random Forest features (Q8), with WaveNum added to capture time effects, giving 12 variables in total. (Age, LifeSatis, GovtRespons, PrivateState, HardWork, Competition, PolLeader, ILPolitics, PolExperts, ILReligion, and PolArmy)

Standardisation (Z-score Scaling): All 11 numeric predictors were z-score standardised (mean 0, SD 1) using ZAF training data parameters, and the same scaling was applied to the test set to avoid data leakage.

Factor to Numeric Output: CWomensMvt (Low/High factor) was modelled using a sigmoid output in *nnet*, with a 0.5 threshold used to classify Low/High.

Network Architecture

Table 23. ANN Architecture Summary

Parameter	Value	Justification
Input nodes	12	11 scaled predictors + WaveNum
Hidden layers	1	Sufficient for binary classification on tabular data

Hidden units	5	Balances model capacity vs. overfitting for n = 548
Output nodes	1	Binary classification using sigmoid activation
Activation	Sigmoid (logistic)	Default in nnet; suitable for binary output
Weight decay	0.01	L2 regularisation to prevent overfitting on small country subset
Max iterations	300	Ensures convergence
Total weights	71	$(12 \times 5) + 5 + (5 \times 1) + 1$
Training data	All 548 South Africa (ZAF) observations	Full country dataset used

Table 24: ANN Metrics - South Africa | CWomensMvt

Metric	Value
Accuracy	0.7791
Precision	0.7727
Recall	0.8854
F1	0.8252

Table 25. Confusion Matrix — ANN (South Africa | CWomensMvt)

	Actual Low	Actual High
Predicted Low	42	11
Predicted High	25	85

The ANN performed strongly on the South Africa test set, achieving an F1-score of 0.8252 and an AUC of 0.795, significantly outperforming the best ensemble models from Question 5 (F1: 0.57–0.73 for CWomensMvt). This indicates that the ANN captured non-linear relationships and feature interactions more effectively in the country-specific data.

Recall was high at 0.8854, correctly identifying most High cases (85 out of 96 actual High confidence cases), though Precision (0.7727) shows some false positives (25 Low cases were incorrectly classified as High), which is reasonable given the small test size (n = 163).

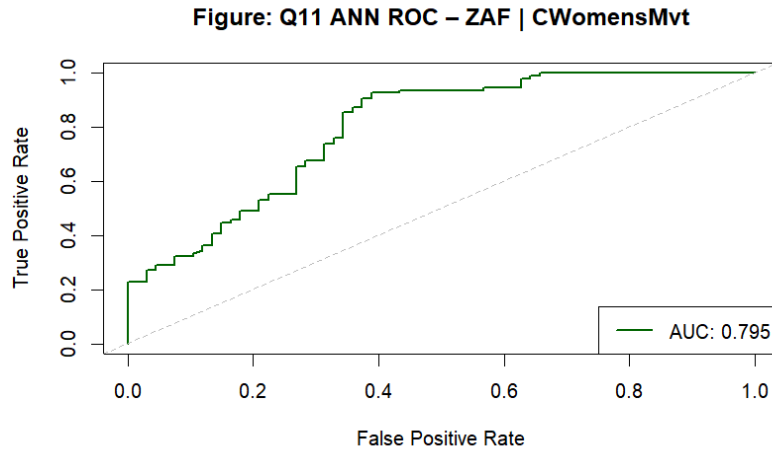


Figure 17: ANN ROC - ZAF | CWomensMvt

The ROC curve shows strong separation between classes, with the AUC of 0.795 confirming good discriminative ability, which is higher than the ensemble AUCs for CWomensMvt (which ranged from approximately 0.57–0.64).

Overall, the ANN's strong performance can be attributed to three factors: the use of country-specific training data that reduces cross-cultural noise present in the full dataset; feature selection that eliminated low-importance predictors identified in Q8; and standardisation which enabled the gradient-based optimisation to converge efficiently.

12. ANN Performance Over Waves

Wave Identification: For South Africa (ZAF), Wave 5 (n = 49) and Wave 6 (n = 43) had the most test observations. The same ANN model from Q11 was applied to each wave without retraining to ensure a fair comparison over time.

Table 26. Confusion Matrices by Wave — ZAF | CWomensMvt

		Wave 5		Wave 6	
		Actual Low	Actual High	Actual Low	Actual High
Predicted	Low	10	3	13	4
Predicted	High	11	25	5	21

Table 27. ANN Performance Comparison by Wave

Wave	N Test	Accuracy	Precision	Recall	F1	AUC	Δ vs Wave 5
Wave 5	49	0.7143	0.6944	0.8929	0.7812	0.7347	-
Wave 6	43	0.7907	0.8077	0.8400	0.8235	0.8056	+0.0423 F1 / +0.0709 AUC

Overall Performance Improvement Across Waves

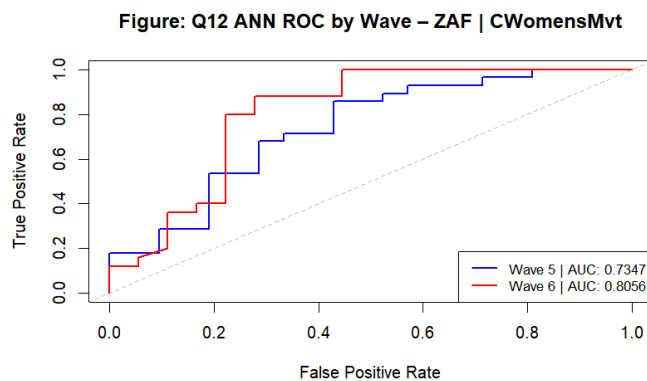


Figure 18. Combined ROC Curves for ANN Classification of CWomensMvt across Wave 5 and Wave 6, ZAF

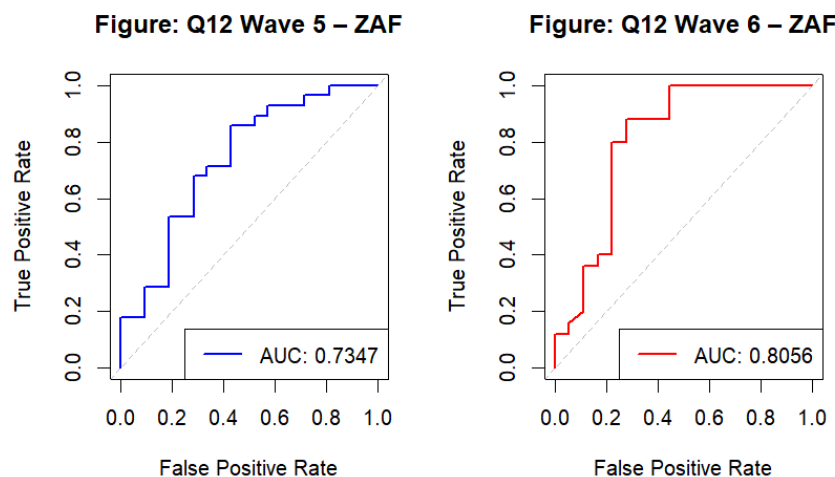


Figure 19. ROC Curves for ANN Classification of CWomensMvt by Wave (a) Wave 5 and (b) Wave 6, ZAF

The ANN shows consistent improvement from Wave 5 to Wave 6 across all metrics. F1 increased from 0.7812 to 0.8235, accuracy from 0.7143 to 0.7907, and AUC from 0.7347 to 0.8056. The ROC curves (Figure 17 & 18) also confirm better separation in Wave 6, with the curve closer to the top-left corner.

Table 28. ANN Classification Performance by Survey Wave - ZAF | CWomensMvt

Wave	N_Test	Accuracy	Precision	Recall	F1	AUC
5	49	0.7143	0.6944	0.8929	0.7812	0.7347
6	43	0.7907	0.8077	0.8400	0.8235	0.8056

Precision vs Recall Trade-off

Wave 5 has a higher recall (0.8929) but a lower precision (0.6944), meaning it captures more true positives but with more false positives. Wave 6 is more balanced, with improved

precision (0.8077) and slightly lower recall (0.8400), resulting in fewer false positives and better overall classification balance.

Temporal Stability and Possible Explanations

- **Stability over time:** F1 remains above 0.78 in both waves, suggesting the predictors of confidence in the Women's Movement stayed consistent across survey periods, demonstrating good temporal generalisation
- **Social or temporal effects:** Wave 6 (2017–2022) coincides with heightened gender equity awareness in South Africa (#MeToo, #TotalShutdown), likely strengthening the relationship between predictors like GovtRespons and ILPolitics with CWomensMvt, improving model discrimination
- **Data drift:** The slight Recall decrease (0.8929 → 0.8400) indicates a minor distributional shift across waves, but the overall performance improvement confirms the ANN captures the core signal robustly despite this drift

Comparison with Q11 Overall Performance

The overall Q11 test set performance (F1 = 0.8252, AUC = 0.795) sits between the two waves. Wave 6 closely matches this overall result, suggesting it reflects the dominant learned patterns, while Wave 5 shows slightly weaker performance, likely due to earlier-period variation or the smaller sample size reducing the stability of evaluation metrics.

Conclusion

Overall, the ANN model trained on South African data exhibits **temporal stability with modest improvement** from Wave 5 to Wave 6, with gains mainly in precision and AUC. These findings indicate that societal attitudes toward the Women's Movement in South Africa became more predictable from the selected WVS attributes over time, potentially reflecting growing social consensus around gender equity in the later wave period.

AI Statement

I used AI in the following ways:

i) To refine my own work and academic language

I submitted the sections I wrote to Grammarly to improve my phrasing, vocabulary, and language accuracy (5 iterations). I further rephrased it to reflect my tone and writing style.

ii) To help clarify concepts related to the assignment

I used ChatGPT to clarify and deepen my understanding of key concepts covered in FIT3152, specifically Decision Tree pruning and Gini impurity, Out-Of-Bag (OOB) error in Random Forest, and ANN architecture, including hidden layers, weights, and sigmoid activation. These clarifications helped me better interpret and articulate my own analytical results, but all analysis, code, and written conclusions were produced independently.

References

Breiman, L. (1996). Bagging predictors. *Machine Learning*, 24(2), 123–140.
<https://doi.org/10.1007/BF00058655>

Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.
<https://doi.org/10.1023/A:1010933404324>

Freund, Y., & Schapire, R. E. (1997). A decision-theoretic generalisation of on-line learning and an application to boosting. *Journal of Computer and System Sciences*, 55(1), 119–139.
<https://doi.org/10.1006/jcss.1997.1504>

Galar, M., Fernández, A., Barrenechea, E., Bustince, H., & Herrera, F. (2012). A review on ensembles for the class imbalance problem. *IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews)*, 42(4), 463–484.
<https://doi.org/10.1109/TSMCC.2011.2170348>

Rish, I. (2001). An empirical study of the Naïve Bayes classifier. In *Proceedings of the IJCAI 2001 Workshop on Empirical Methods in Artificial Intelligence*, 3(22), 41–46.

FIT3152_A2

```
library(ROCR)
```

Warning: package 'ROCR' was built under R version 4.5.3

```
library(rpart)          # Decision Tree
library(rpart.plot)     # Tree visualisation
```

Warning: package 'rpart.plot' was built under R version 4.5.3

```
library(e1071)          # Naive Bayes
```

Warning: package 'e1071' was built under R version 4.5.3

```
library(ipred)          # Bagging
```

Warning: package 'ipred' was built under R version 4.5.3

```
library(adabag)         # Boosting
```

Warning: package 'adabag' was built under R version 4.5.3

Loading required package: caret

Warning: package 'caret' was built under R version 4.5.3

Loading required package: ggplot2

Attaching package: 'ggplot2'

The following object is masked from 'package:e1071':

element

Loading required package: lattice

Loading required package: foreach

Warning: package 'foreach' was built under R version 4.5.3

Loading required package: doParallel

Warning: package 'doParallel' was built under R version 4.5.3

Loading required package: iterators

Loading required package: iterators

Warning: package 'iterators' was built under R version 4.5.3

Loading required package: parallel

Attaching package: 'adabag'

The following object is masked from 'package:ipred':

bagging

```
library(randomForest) # Random Forest
```

Warning: package 'randomForest' was built under R version 4.5.3

randomForest 4.7-1.2

Type rfNews() to see new features/changes/bug fixes.

Attaching package: 'randomForest'

The following object is masked from 'package:ggplot2':

margin

```
library(pROC) # ROC / AUC
```

Warning: package 'pROC' was built under R version 4.5.3

Type 'citation("pROC")' for a citation.

Attaching package: 'pROC'

The following objects are masked from 'package:stats':

cov, smooth, var

```
library(caret) # confusionMatrix, cross-validation
library(nnet) # Artificial Neural Network
library(dplyr) # Data wrangling
```

Attaching package: 'dplyr'

The following object is masked from 'package:randomForest':

combine

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)      # Plotting
library(knitr)         # kable() tables
library(kableExtra)    # kable styling (striped rows, bold columns etc.)
```

Warning: package 'kableExtra' was built under R version 4.5.3

Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

group_rows

```
library(tidyr)         # pivot_longer for long-format plotting
```

```
# =====
# DATA CREATION
# =====
rm(list = ls())
set.seed(33986010) # Your Student ID is the random seed
WD = read.csv("WVSBinaryExtract.csv")
selected_cols = c(sample(3:49, 30), sample(50:63, 3))
WD = WD[c(1:2, selected_cols)]
WD = WD[sample(nrow(WD), 20000, replace = FALSE),]

cat("Dimensions:", dim(WD), "\n")
```

Dimensions: 20000 35

```
cat("Columns:", paste(names(WD), collapse = ", "), "\n")
```

Columns: Country, Wave, PolArmy, ILPolitics, ILWork, HardWork, ICQTolerance, ILReligion, LifeSatis, ICQDetermination, PolBoycott, ICQHardWork, ICQIndependence, PolExperts, ICQThrift, ACTArtsEd, FutureWork, ILLeisure, PolLeader, Competition, ACTSport, ACTProfessional, PolDemons, ACTPolParty, PrivateState, GovtRespons, PolDemoc, ACTHumanitarian, ICQObedience, Age, ILFriends, ICQFaith, CWomensMvt, CGovernment, CUnions

```
# Last 3 columns = binary class variables (sampled from cols 50-63)
class_vars <- tail(names(WD), 3)
cat("Class variables:", paste(class_vars, collapse = ", "), "\n")
```

Class variables: CWomensMvt, CGovernment, CUnions

```
# =====
# QUESTION 1 - Data Exploration
# =====
cat("\n==== Q1: Data Exploration =====\n")
```

==== Q1: Data Exploration =====

```
# -----
# 1a. Class variable proportions
# -----
prop_rows <- list()

for (cv in class_vars) {
  tbl <- table(factor(WD[[cv]], levels = c(0, 1)))
  prop <- prop.table(tbl)

  ratio <- if (min(prop) == 0) NA else max(prop) / min(prop)

  prop_rows[[cv]] <- data.frame(
    Class_Variable = cv,
    N_High         = as.integer(tbl["1"]),
    N_Low          = as.integer(tbl["0"]),
    Prop_High      = round(as.numeric(prop["1"]), 4),
    Prop_Low       = round(as.numeric(prop["0"]), 4),
    ImbalRatio     = round(ratio, 3)
  )
}

prop_df <- do.call(rbind, prop_rows)
rownames(prop_df) <- NULL

kable(prop_df,
      caption = "Table 1. Class variable proportions (High = 1, Low = 0)",
      col.names = c("Class Variable", "Number of High", "Number of Low",
                    "Proportions High", "Proportions Low", "Imbalance Ratio")) %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                full_width = FALSE)
```

Table 1. Class variable proportions (High = 1, Low = 0)

Class Variable	Number of High	Number of Low	Proportions High	Proportions Low	Imbalance Ratio
----------------	----------------	---------------	------------------	-----------------	-----------------

CWomensMvt	9160	6418	0.5880	0.4120	1.427
CGovernment	8067	9267	0.4654	0.5346	1.149
CUnions	7162	10349	0.4090	0.5910	1.445

```
# -----
# 1b. Predictor summary statistics (kable replacing summary())
# -----
predictors <- WD[, 3:(ncol(WD) - 3)]
pred_cols <- names(predictors)

stats_df <- data.frame(
  Variable = pred_cols,
  Mean     = round(sapply(predictors, mean, na.rm = TRUE), 3),
  SD       = round(sapply(predictors, sd, na.rm = TRUE), 3),
  Min      = round(sapply(predictors, min, na.rm = TRUE), 3),
  Median   = round(sapply(predictors, median, na.rm = TRUE), 3),
  Max      = round(sapply(predictors, max, na.rm = TRUE), 3),
  N_Missing = as.integer(colSums(is.na(predictors))),
  N_Negative = as.integer(sapply(predictors,
                                function(x) sum(x < 0, na.rm = TRUE)))
)
rownames(stats_df) <- NULL

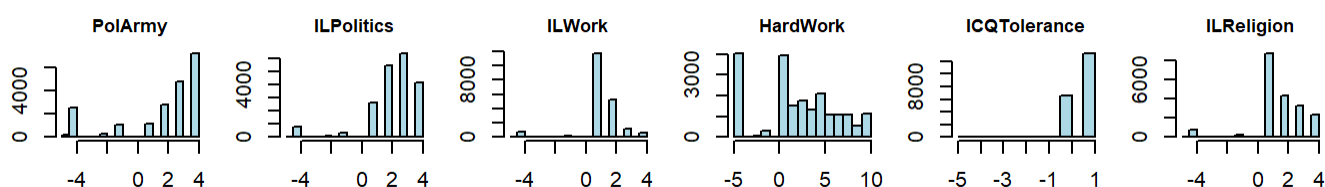
kable(stats_df,
  caption = paste("Table 2. Summary statistics of predictor attributes",
    "(N =", nrow(WD), "). N_Negative indicates WVS",
    "system-missing codes (-1 to -5) requiring removal."),
  col.names = c("Variable", "Mean", "SD", "Min",
    "Median", "Max", "N Missing", "N Negative"),
  row.names = FALSE) |>
kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
  full_width = TRUE) |>
column_spec(8, bold = TRUE,
  color = ifelse(stats_df$N_Negative > 0, "red", "black"))
```

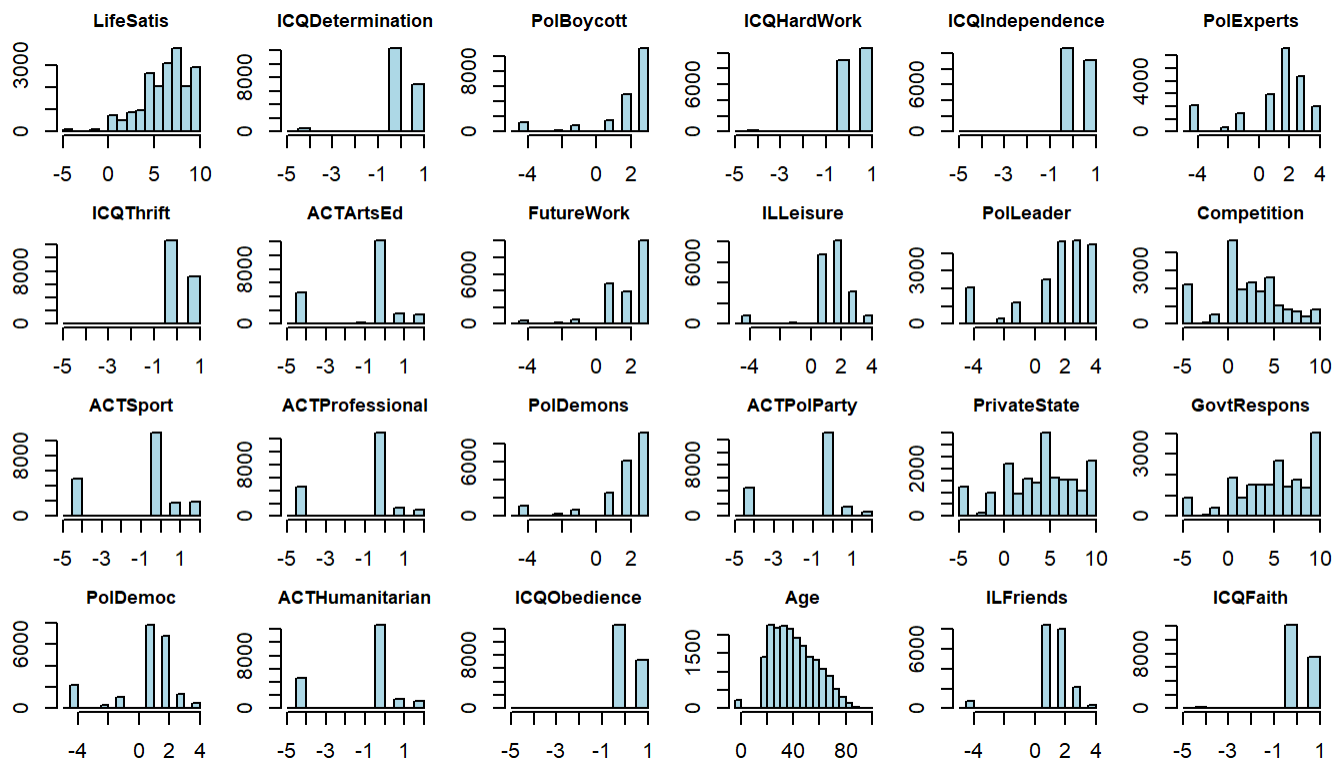
Table 2. Summary statistics of predictor attributes (N = 20000). N_Negative indicates WVS system-missing codes (-1 to -5) requiring removal.

Variable	Mean	SD	Min	Median	Max	N Missing	N Negative
PolArmy	1.862	2.727	-5	3	4	0	4074
ILPolitics	2.301	1.702	-5	3	4	0	1217
ILWork	1.230	1.394	-5	1	4	0	1123
HardWork	2.521	4.224	-5	3	10	0	4426
ICQTolerance	0.643	0.580	-5	1	1	0	114
ILReligion	1.661	1.635	-5	2	4	0	1170

LifeSatis	6.657	2.616	-5	7	10	0	285
ICQDetermination	0.237	0.861	-5	0	1	0	582
PolBoycott	1.933	1.851	-5	3	3	0	2338
ICQHardWork	0.487	0.689	-5	1	1	0	235
ICQIndependence	0.435	0.596	-5	0	1	0	132
PolExperts	1.330	2.275	-5	2	4	0	4021
ICQThrift	0.335	0.577	-5	0	1	0	137
ACTArtsEd	-0.718	1.877	-5	0	2	0	4833
FutureWork	2.018	1.398	-5	3	3	0	1095
ILLeisure	1.613	1.454	-5	2	4	0	1064
PolLeader	1.677	2.402	-5	2	4	0	3648
Competition	2.775	3.491	-5	3	10	0	2829
ACTSport	-0.732	1.993	-5	0	2	0	5136
ACTProfessional	-0.762	1.837	-5	0	2	0	4716
PolDemons	1.831	1.780	-5	2	3	0	2084
ACTPolParty	-0.760	1.807	-5	0	2	0	4620
PrivateState	4.552	3.783	-5	5	10	0	2351
GovtRespons	5.530	3.746	-5	6	10	0	1406
PolDemoc	0.858	1.973	-5	1	4	0	3510
ACTHumanitarian	-0.753	1.858	-5	0	2	0	4760
ICQObedience	0.340	0.579	-5	0	1	0	142
Age	41.268	16.891	-5	39	99	0	229
ILFriends	1.460	1.359	-5	2	4	0	928
ICQFaith	0.324	0.679	-5	0	1	0	268

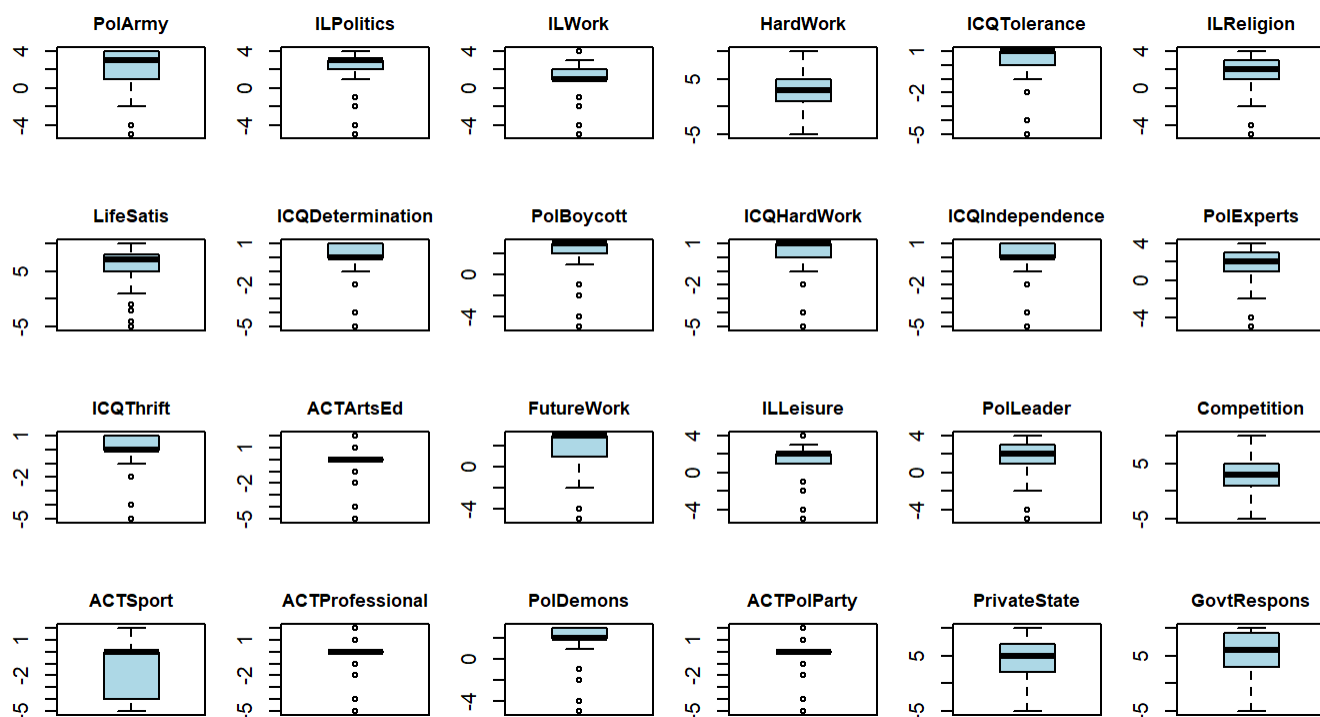
```
# Histograms for all numeric predictors (4 per page)
num_pred_names <- names(predictors)
n_preds <- length(num_pred_names)
par(mfrow=c(5,6), mar=c(2,2,2,1))
for (nm in num_pred_names) {
  hist(predictors[[nm]], main=nm, xlab="", col="lightblue",
        breaks=15, cex.main=0.85)
}
```

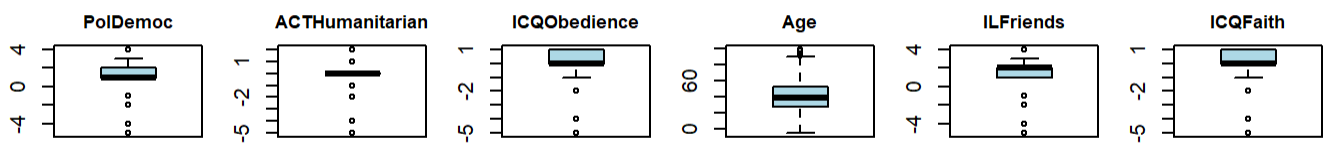




```
par(mfrow=c(1,1))

# Box plots for all numeric predictors
par(mfrow=c(5,6), mar=c(2,2,2,1))
for (nm in num_pred_names) {
  boxplot(predictors[[nm]], main=nm, xlab="", col="lightblue",
          cex.main=0.85, outline=TRUE)
}
```





```
par(mfrow=c(1,1))

# -----
# 1c. NA and Negative Value Summary
# -----
# Count NAs (including negative-coded WVS values)
WD_raw_check <- WD # before pre-processing negatives are already in the data
neg_na_counts <- sapply(WD[pred_cols], function(x) sum(x < 0 | is.na(x), na.rm = TRUE))
pct_missing <- round(neg_na_counts / nrow(WD) * 100, 2)

missing_df <- data.frame(
  Attribute      = names(neg_na_counts),
  Missing_Count  = as.vector(neg_na_counts),
  Missing_Pct    = as.vector(pct_missing)
)
missing_df <- missing_df[order(-missing_df$Missing_Pct), ]

cat("\n--- Missing / Negative-Coded Values per Predictor ---\n")
```

--- Missing / Negative-Coded Values per Predictor ---

```
kable(missing_df,
      caption = "Table 1. Missing and negative-coded values per predictor attribute",
      row.names = FALSE,
      format = "html") %>%
  kable_styling(
    bootstrap_options = c("striped", "hover", "condensed", "responsive"),
    full_width = FALSE,
    position = "center",
    font_size = 14
  ) %>%
  row_spec(0, bold = TRUE, background = "#4472C4", color = "white") %>%
  column_spec(1, bold = TRUE, width = "20em") %>%
  scroll_box(width = "100%")
```

Table 1. Missing and negative-coded values per predictor attribute

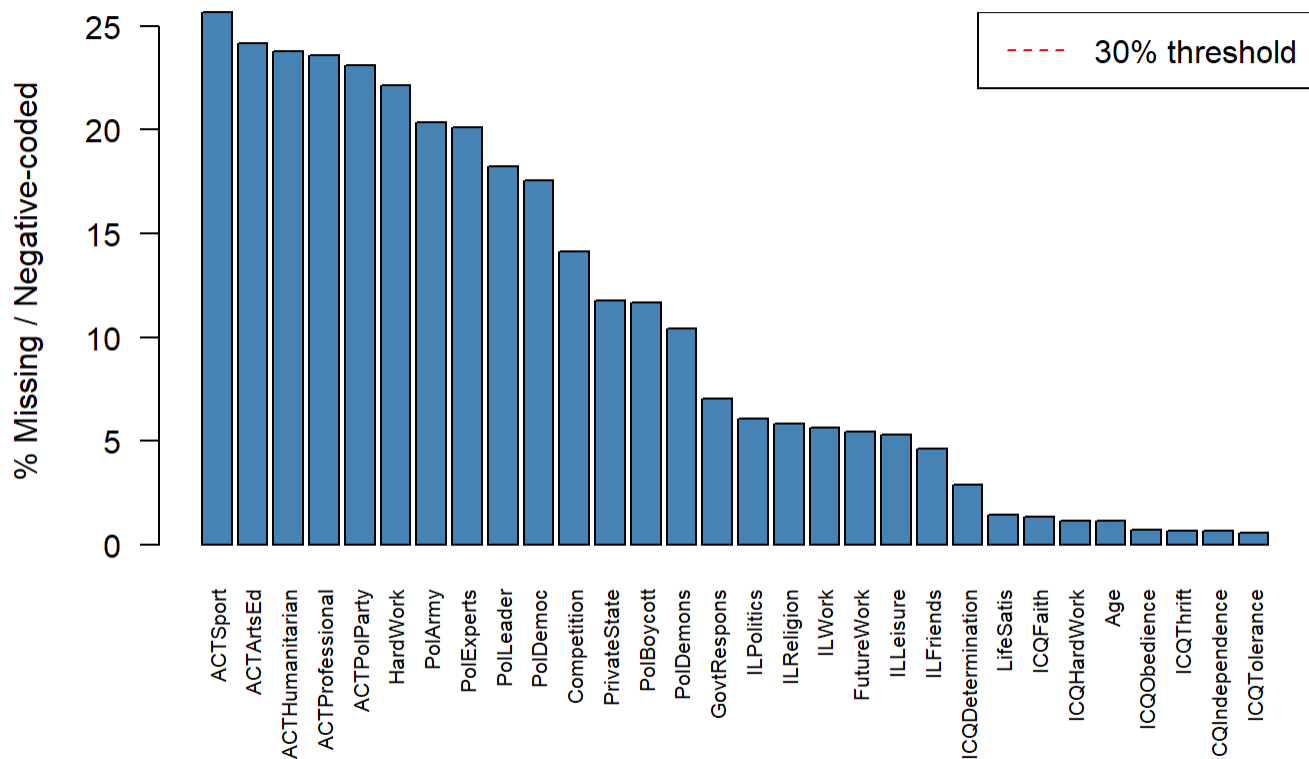
Attribute	Missing_Count	Missing_Pct
ACTSport	5136	25.68

ACTReport	5155	23.88
ACTArtsEd	4833	24.16
ACTHumanitarian	4760	23.80
ACTProfessional	4716	23.58
ACTPolParty	4620	23.10
HardWork	4426	22.13
PolArmy	4074	20.37
PolExperts	4021	20.10
PolLeader	3648	18.24
PolDemoc	3510	17.55
Competition	2829	14.14
PrivateState	2351	11.76
PolBoycott	2338	11.69
PolDemons	2084	10.42
GovtRespons	1406	7.03
ILPolitics	1217	6.08
ILReligion	1170	5.85
ILWork	1123	5.62
FutureWork	1095	5.47
ILLeisure	1064	5.32
ILFriends	928	4.64
ICQDetermination	582	2.91
LifeSatis	285	1.43
ICQFaith	268	1.34
ICQHardWork	235	1.18
Age	229	1.15
ICQObedience	142	0.71
ICQThrift	137	0.69
ICQIndependence	132	0.66
ICQTolerance	114	0.57

```
# Bar chart of % missing
par(mar = c(8, 4, 3, 1))
barplot(missing_df$Missing_Pct,
        names.arg = missing_df$Attribute,
        las      = 2,
        col      = ifelse(missing_df$Missing_Pct > 30, "tomato", "steelblue"),
        ...)
```

```
ylab      = "% Missing / Negative-coded",
main      = "Figure. Percentage of missing values per predictor",
cex.names = 0.65)
abline(h = 30, lty = 2, col = "red")
legend("topright", legend = "30% threshold", lty = 2, col = "red")
```

Figure. Percentage of missing values per predictor



```
par(mar = c(5, 4, 4, 2))
```

```
# =====
# Q1: Country and Wave observation counts
# =====

# --- Country counts ---
country_counts <- sort(table(WD$Country), decreasing = TRUE)

country_df <- data.frame(
  Country    = names(country_counts),
  N_Obs      = as.integer(country_counts),
  Percentage = round(as.integer(country_counts) / nrow(WD) * 100, 2)
)
rownames(country_df) <- NULL
```

```
kable(country_dt,  
      caption = "Table 3. Number of observations per country",  
      col.names = c("Country", "N Observations", "% of Total")) %>%  
kable_styling(bootstrap_options = c("striped", "hover", "condensed"),  
              full_width = FALSE)
```

Table 3. Number of observations per country

Country	N Observations	% of Total
ZAF	900	4.50
IND	776	3.88
COL	700	3.50
MEX	661	3.31
TUR	654	3.27
CHN	587	2.94
RUS	560	2.80
JPN	525	2.62
USA	499	2.50
EGY	498	2.49
KOR	487	2.44
CAN	478	2.39
AUS	457	2.28
NGA	442	2.21
DEU	421	2.10
BRA	420	2.10
ARG	407	2.04
CHL	405	2.02
IRQ	388	1.94
PER	387	1.93
ESP	372	1.86
IRN	364	1.82
IDN	352	1.76
PAK	344	1.72
ROU	332	1.66
UKR	328	1.64
SGP	307	1.54
JOR	296	1.48
MAR	279	1.40

MDA	163	0.81
THA	272	1.36
BLR	271	1.35
ARM	265	1.32
GEO	264	1.32
NLD	264	1.32
GBR	263	1.31
TWN	261	1.31
URY	255	1.27
PHL	254	1.27
SRB	245	1.23
POL	244	1.22
CHE	237	1.19
HKG	228	1.14
VNM	225	1.12
BGD	221	1.10
KGZ	221	1.10
NZL	220	1.10
MYS	212	1.06
ZWE	212	1.06
SWE	211	1.05
VEN	203	1.01
HUN	186	0.93
CZE	170	0.85
CYP	168	0.84
SVN	165	0.83
PRI	164	0.82
MDA	163	0.81
SVK	162	0.81
FIN	148	0.74

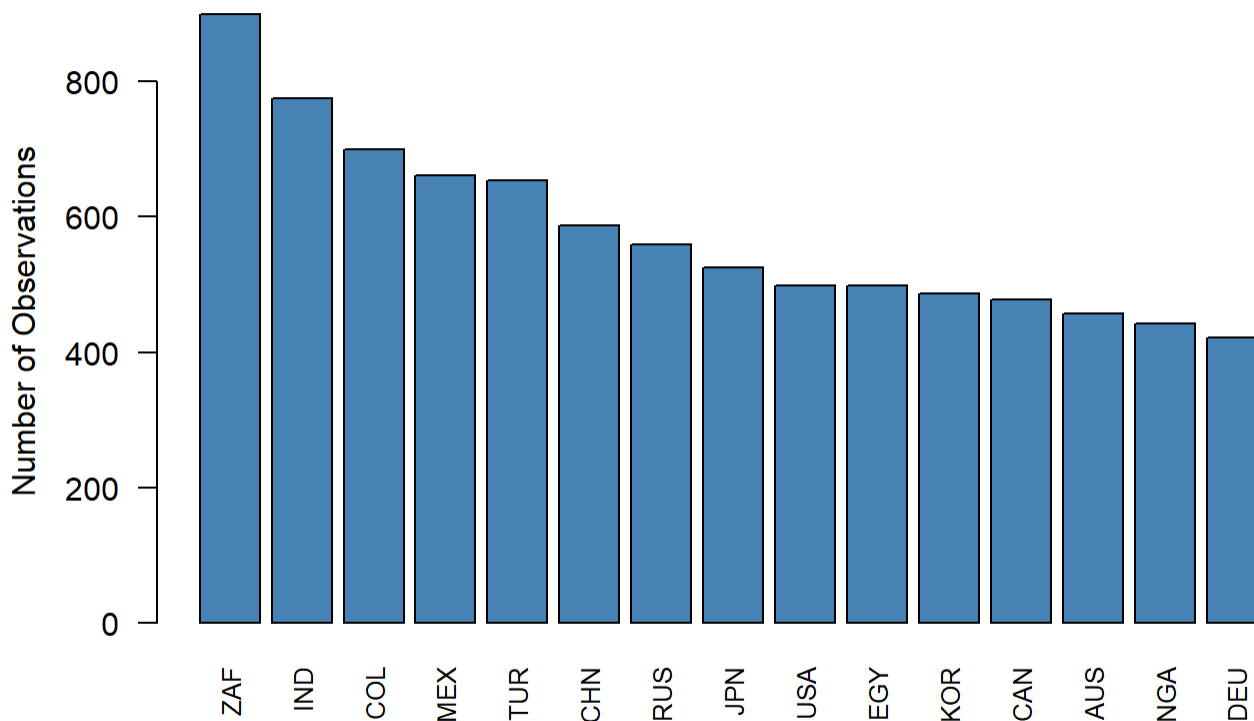
```
# Bar chart - top 15 countries
top15 <- head(country_df, 15)
par(mar = c(6, 4, 3, 1))
barplot(top15$N_Obs,
        names.arg = top15$Country,
        las       = 2,
        col       = "steelblue".
```

```

---
ylab      = "Number of Observations",
main      = "Figure: Top 15 Countries by Observation Count",
cex.names = 0.8)

```

Figure: Top 15 Countries by Observation Count



```

par(mar = c(5, 4, 4, 2))

# --- Wave counts ---
wave_counts <- sort(table(WD$Wave), decreasing = TRUE)

wave_df <- data.frame(
  Wave      = names(wave_counts),
  N_Obs     = as.integer(wave_counts),
  Percentage = round(as.integer(wave_counts) / nrow(WD) * 100, 2)
)
rownames(wave_df) <- NULL

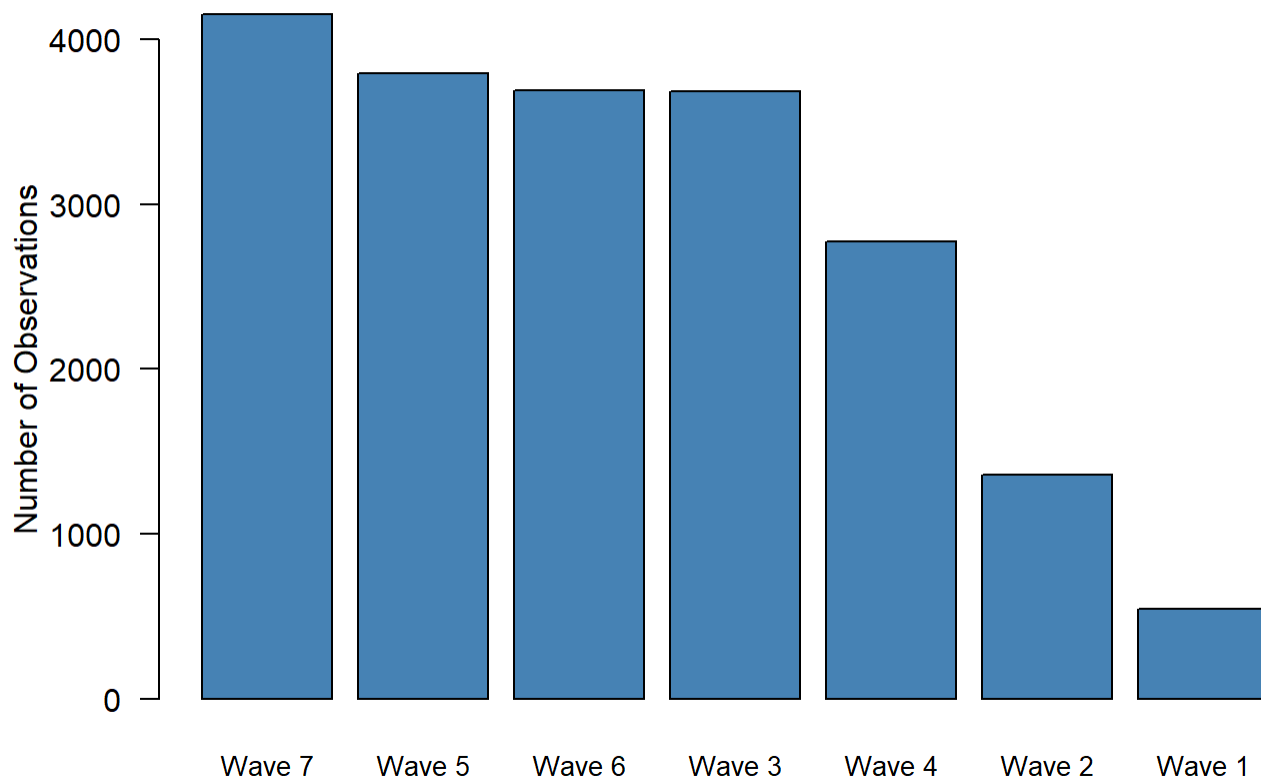
kable(wave_df,
      caption = "Table 4. Number of observations per survey wave",
      col.names = c("Wave", "N Observations", "% of Total")) %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                full_width = FALSE)

```

Table 4. Number of observations per survey wave

Wave	N Observations	% of Total
7	4153	20.76
5	3795	18.98
6	3688	18.44
3	3682	18.41
4	2774	13.87
2	1362	6.81
1	546	2.73

```
# Bar chart - wave distribution
par(mar = c(4, 4, 3, 1))
barplot(wave_df$N_Obs,
        names.arg = paste("Wave", wave_df$Wave),
        las       = 1,
        col       = "steelblue",
        ylab      = "Number of Observations",
        main      = "Figure: Observations per Survey Wave",
        cex.names = 0.85)
```

Figure: Observations per Survey Wave

```

par(mar = c(5, 4, 4, 2))

# --- Combined country + wave heatmap ---
# Shows which countries appear in which waves
cw_table <- table(WD$Country, WD$Wave)
cw_df     <- as.data.frame.matrix(cw_table)

kable(cw_df,
      caption = "Table 5. Observation counts by Country and Wave",
      format  = "html") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                full_width = TRUE,
                font_size  = 11) %>%
  scroll_box(width = "100%", height = "400px")

```

	1	2	3	4	5	6	7
ARG	60	51	66	65	65	53	47
ARM	0	0	113	0	0	72	80
AUS	57	0	118	0	89	84	109
BGD	0	0	71	96	0	0	54
BLR	0	47	139	0	0	85	0
BRA	0	96	69	0	74	93	88
CAN	0	0	0	107	124	0	247
CHE	0	81	82	0	74	0	0
CHL	0	105	53	70	61	59	57
CHN	0	42	97	53	112	121	162
COL	0	0	341	0	194	70	95
CYP	0	0	0	0	66	58	44

```

# =====
# QUESTION 2 - Pre-processing
# =====
cat("\n==== Q2: Pre-processing =====\n")

```

==== Q2: Pre-processing =====

```

WD_clean <- WD
pred_cols <- names(WD_clean)[3:(ncol(WD_clean) - 3)]

# Replace WVS negative system-missing codes with NA
for (col in pred_cols) {
  if (is.numeric(WD_clean[[col]])) {
    WD_clean[[col]][WD_clean[[col]] < 0] <- NA
  }
}

# Remove rows where any class variable is NA
WD_clean <- WD_clean[complete.cases(WD_clean[, class_vars]), ]

# Impute remaining predictor NAs with column median
for (col in pred_cols) {
  if (is.numeric(WD_clean[[col]])) {
    med <- median(WD_clean[[col]], na.rm = TRUE)
    WD_clean[[col]][is.na(WD_clean[[col]])] <- med
  }
}

# Convert class variables to factors (0 = Low, 1 = High)
for (cv in class_vars) {
  WD_clean[[cv]] <- factor(WD_clean[[cv]], levels = c(0, 1),
                           labels = c("Low", "High"))
  cat(sprintf(" %s -> factor levels: %s\n", cv,
          paste(levels(WD[[cv]]), collapse = ", ")))
}

```

```

CWomensMvt -> factor levels:
CGovernment -> factor levels:
CUnions -> factor levels:

```

```

# Convert Country and Wave to factors (categorical identifiers)
WD_clean$Country <- as.factor(WD_clean$Country)
WD_clean$Wave <- as.factor(WD_clean$Wave)

# Scaling / Standardisation
cat("\n--- 2f. Scaling note ---\n")

```

```

--- 2f. Scaling note ---

```

```

cat("  Scaling deferred to Q11 (ANN only).\n")

```

```

Scaling deferred to Q11 (ANN only).

```

```

cat("  Tree-based models and Naive Bayes are scale-invariant.\n")

```

Tree-based models and Naive Bayes are scale-invariant.

```
# Final Dataset Overview
```

```
cat("\n--- 2g. Final dataset dimensions after pre-processing ---\n")
```

```
--- 2g. Final dataset dimensions after pre-processing ---
```

```
final_summary <- data.frame(
  Item = c("Total observations", "Predictor columns",
           "Convert class variables to factors (High/Low)", "Remaining NAs"),
  Value = c(nrow(WD_clean), length(pred_cols), paste(paste(class_vars, collapse = ", "), "converted"),
            0)
)
print(kable(final_summary,
            caption = "Table 6. Dataset summary after pre-processing",
            row.names = FALSE))
```

Table: Table 6. Dataset summary after pre-processing

Item	Value
Total observations	14112
Predictor columns	30
Convert class variables to factors (High/Low)	CWomensMvt, CGovernment, CUnions converted
Remaining NAs	0

```
# =====
# QUESTION 3: Train/Test Split (70/30)
# =====
cat("\n===== QUESTION 3: TRAIN/TEST SPLIT =====\n")
```

```
===== QUESTION 3: TRAIN/TEST SPLIT =====
```

```
set.seed(33986010)
train.row <- sample(1:nrow(WD_clean), 0.7 * nrow(WD_clean))
WD.train <- WD_clean[train.row, ]
WD.test <- WD_clean[-train.row, ]

cat("Training set size:", nrow(WD.train), "\n")
```

Training set size: 9878

```
cat("Test set size: ", nrow(WD.test), "\n")
```

Test set size: 4234

```
# Define predictor columns (exclude Country, Wave, and all class vars)
all_cols      <- names(WD)
non_pred      <- c("Country", "Wave", class_vars)
pred_cols     <- setdiff(all_cols, non_pred)
cat("Predictor columns used:", length(pred_cols), "\n")
```

Predictor columns used: 30

```
# =====
# HELPER FUNCTIONS
# =====

# Build formula: class_var ~ predictors
make_formula <- function(cv, preds) {
  as.formula(paste(cv, "~", paste(preds, collapse = " + ")))
}

# Compute metrics from confusion matrix
get_metrics <- function(cm) {
  # cm is a 2x2 table: rows=predicted, cols=actual; labels Low/High
  # Positive class = "High"
  tp <- cm["High", "High"]
  tn <- cm["Low", "Low"]
  fp <- cm["High", "Low"]
  fn <- cm["Low", "High"]
  acc <- (tp + tn) / sum(cm)
  prec <- ifelse((tp + fp) == 0, 0, tp / (tp + fp))
  rec <- ifelse((tp + fn) == 0, 0, tp / (tp + fn))
  f1 <- ifelse((prec + rec) == 0, 0, 2 * prec * rec / (prec + rec))
  c(Accuracy=round(acc,4), Precision=round(prec,4),
    Recall=round(rec,4), F1=round(f1,4))
}
```

```
# =====
# QUESTION 4: Fit Models on Training Data
# =====
cat("\n===== QUESTION 4: MODEL FITTING =====\n")
```

===== QUESTION 4: MODEL FITTING =====

```
model_names <- c("Decision Tree", "Naive Bayes", "Bagging",
  "Boosting", "Random Forest")

prob_list <- list()
```

```

for (cv in class_vars) {
  cat("\n----- Class Variable:", cv, "-----\n")
  fml <- make_formula(cv, pred_cols)

  # ---- 4a. Decision Tree with cross-validation pruning ----
  cat(" Fitting Decision Tree...\n")
  dt.model <- rpart(fml, data = WD.train, method = "class",
                    control = rpart.control(xval = 10)) # 10-fold cross-validation

  # Prune using optimal CP from cross-validation
  best_cp <- dt.model$cptable[which.min(dt.model$cptable[, "xerror"]), "CP"]
  dt.model <- prune(dt.model, cp = best_cp)

  if (!is.null(dt.model$splits) && nrow(dt.model$splits) > 0) {
    # extra = 106: shows predicted class + % of observations in node + n
    # type = 0: simple lines like plain plot(), no boxes
    rpart.plot(dt.model,
               type = 0,
               extra = 106,
               under = TRUE,
               cex = 0.75,
               digits = 2,
               roundint = FALSE, # stops rounding to nearest integer
               main = paste("Figure: Decision Tree -", cv),
               box.palette = "auto",
               shadow.col = "gray")
  } else {
    cat(" Note: single-node tree for", cv,
        "\n- no splits produced (class imbalance). Only CGovernment is plotted.\n")
  }

  # ---- 4b. Naive Bayes ----
  cat(" Fitting Naive Bayes...\n")
  nb.model <- naiveBayes(fml, data = WD.train)
  # Print number of predictor conditional tables as evidence of fitting
  cat(" Naive Bayes fitted with", length(nb.model$tables), "predictor distributions.\n")

  # ---- 4c. Bagging ----
  cat(" Fitting Bagging...\n")
  bag.model <- adabag::bagging(fml, data = WD.train, mfinal = 25)
  cat(" Bagging fitted with", bag.model$terms |> length(), "terms,",
      length(bag.model$trees), "trees.\n")

  # ---- 4d. Boosting ----
  cat(" Fitting Boosting...\n")
  bst.model <- adabag::boosting(fml, data = WD.train, mfinal = 25)
  cat(" Boosting fitted with", length(bst.model$trees), "trees.\n")

  # ---- 4e. Random Forest ----

```

```

cat("  Fitting Random Forest...\n")
rf.model <- randomForest(fml, data = WD.train, ntree = 100, importance = TRUE)
print(rf.model)  # prints OOB error, class error table – useful evidence

# Store models
assign(paste0("dt_", cv), dt.model)
assign(paste0("nb_", cv), nb.model)
assign(paste0("bag_", cv), bag.model)
assign(paste0("bst_", cv), bst.model)
assign(paste0("rf_", cv), rf.model)

# Store predicted probabilities for Q6
bag.raw <- adabag::predict.bagging(bag.model, newdata = WD.test)
bst.raw <- adabag::predict.boosting(bst.model, newdata = WD.test)

prob_list[[cv]] <- list(
  "Decision Tree" = predict(dt.model, WD.test, type = "prob")[, "High"],
  "Naive Bayes"   = predict(nb.model, WD.test, type = "raw")[, "High"],
  "Bagging"       = bag.raw$prob[, 2],
  "Boosting"      = bst.raw$prob[, 2],
  "Random Forest" = predict(rf.model, WD.test, type = "prob")[, "High"]
)

cat("  All models fitted for", cv, "\n")
}

```

----- Class Variable: CWomensMvt -----

Fitting Decision Tree...

Note: single-node tree for CWomensMvt - no splits produced (class imbalance). Only CGovernment is plotted.

Fitting Naive Bayes...

Naive Bayes fitted with 30 predictor distributions.

Fitting Bagging...

Bagging fitted with 3 terms, 25 trees.

Fitting Boosting...

Boosting fitted with 25 trees.

Fitting Random Forest...

Call:

```
randomForest(formula = fml, data = WD.train, ntree = 100, importance = TRUE)
```

Type of random forest: classification

Number of trees: 100

No. of variables tried at each split: 5

OOB estimate of error rate: 41.57%

Confusion matrix:

Low High class.error

Low 1281 2824 0.6879415

High 1282 4491 0.7220682

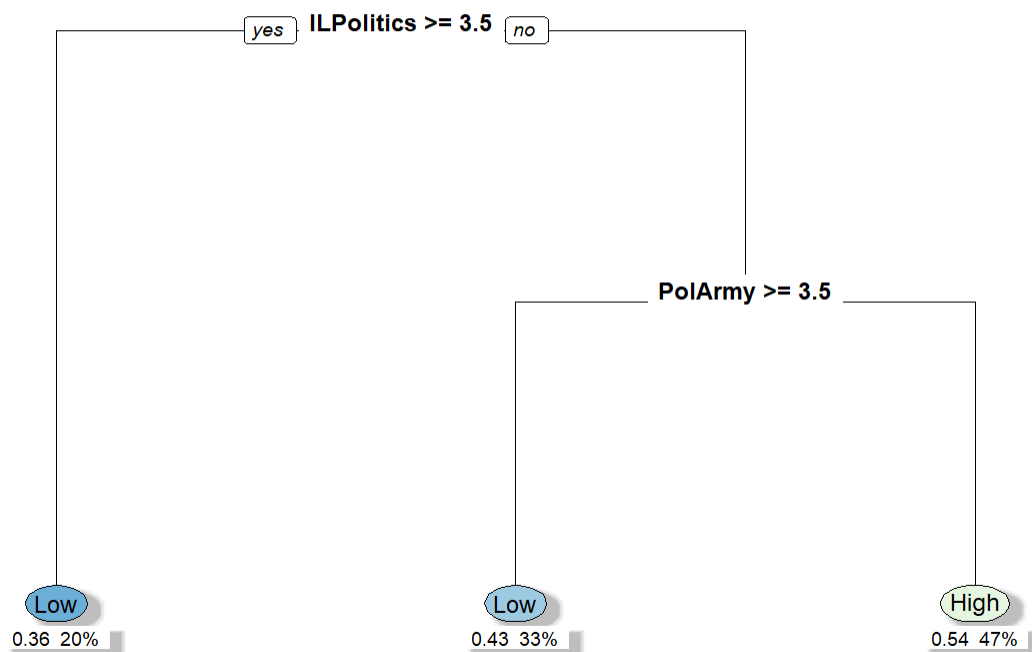
```
High 1202 4751 0.2220002
```

```
All models fitted for CWomensMvt
```

```
----- Class Variable: CGovernment -----
```

```
Fitting Decision Tree...
```

Figure: Decision Tree - CGovernment



```
Fitting Naive Bayes...
```

```
Naive Bayes fitted with 30 predictor distributions.
```

```
Fitting Bagging...
```

```
Bagging fitted with 3 terms, 25 trees.
```

```
Fitting Boosting...
```

```
Boosting fitted with 25 trees.
```

```
Fitting Random Forest...
```

```
Call:
```

```
randomForest(formula = fml, data = WD.train, ntree = 100, importance = TRUE)
```

```
  Type of random forest: classification
```

```
    Number of trees: 100
```

```
No. of variables tried at each split: 5
```

```
  OOB estimate of  error rate: 43.09%
```

```
Confusion matrix:
```

```
  Low High class.error
```

```
Low  3477 1811  0.3424735
```

```
High  2445 2145  0.5226707
```

```

# High 2445 2145 0.5520/9/

```

```

All models fitted for CGovernment

```

```

----- Class Variable: CUnions -----

```

```

Fitting Decision Tree...

```

```

Note: single-node tree for CUnions - no splits produced (class imbalance). Only CGovernment
is plotted.

```

```

Fitting Naive Bayes...

```

```

Naive Bayes fitted with 30 predictor distributions.

```

```

Fitting Bagging...

```

```

Bagging fitted with 3 terms, 25 trees.

```

```

Fitting Boosting...

```

```

Boosting fitted with 25 trees.

```

```

Fitting Random Forest...

```

```

Call:

```

```

randomForest(formula = fml, data = WD.train, ntree = 100, importance = TRUE)

```

```

Type of random forest: classification

```

```

Number of trees: 100

```

```

No. of variables tried at each split: 5

```

```

OOB estimate of error rate: 41.75%

```

```

Confusion matrix:

```

```

Low High class.error

```

```

Low 4512 1226 0.2136633

```

```

High 2898 1242 0.7000000

```

```

All models fitted for CUnions

```

```

# =====
# QUESTION 5: MODEL EVALUATION - Grouped Confusion Matrices
# =====
cat("\n===== QUESTION 5: MODEL EVALUATION =====\n")

```

```

===== QUESTION 5: MODEL EVALUATION =====

```

```

results_list <- list()

for (cv in class_vars) {
  cat("\n----- Class Variable:", cv, "-----\n")

  actual <- WD.test[[cv]]

  # Generate predictions
  dt.pred <- predict(get(paste0("dt_", cv)), WD.test, type = "class")
  nb.pred <- predict(get(paste0("nb_", cv)), WD.test)
  bag.pred <- factor(predict.bagging(get(paste0("bag_", cv)),
                                   newdata = WD.test)$class,
                    levels = c("Low", "High"))
  bst.pred <- factor(predict.boosting(get(paste0("bst_", cv)),

```

```

bst.pred <- factor(predict(boosting(get(paste0("bst_", cv)),
                                   newdata = WD.test)$class,
                                   levels = c("Low", "High")))
rf.pred <- predict(get(paste0("rf_", cv)), WD.test)

# Build confusion matrices
cms <- list(
  "Decision Tree" = table(Predicted = dt.pred, Actual = actual),
  "Naive Bayes"   = table(Predicted = nb.pred, Actual = actual),
  "Bagging"       = table(Predicted = bag.pred, Actual = actual),
  "Boosting"      = table(Predicted = bst.pred, Actual = actual),
  "Random Forest" = table(Predicted = rf.pred, Actual = actual)
)

# Build 1 wide table with all 5 confusion matrices side by side ---
cm_wide <- data.frame(Predicted = c("Low", "High"))

for (mn in names(cms)) {
  cm <- as.data.frame.matrix(cms[[mn]])
  # Add Low and High columns with model name prefix
  cm_wide[[paste0(mn, "_Low")]] <- c(cm["Low", "Low"], cm["High", "Low"])
  cm_wide[[paste0(mn, "_High")]] <- c(cm["Low", "High"], cm["High", "High"])
}

# Build header: Predicted | DT: Low High | NB: Low High | ...
n_models <- length(cms)
col_names <- c("Predicted",
               unlist(lapply(names(cms), function(mn) c("Low", "High"))))

header_top <- c(" " = 1)
for (mn in names(cms)) header_top[mn] <- 2

cat("\n### Confusion Matrices (All Models) -", cv, "\n")
print(
  kable(cm_wide,
        col.names = col_names,
        caption = paste("Confusion Matrices for all models -", cv),
        align = c("l", rep("c", n_models * 2))) %>%
    kable_styling(
      bootstrap_options = c("striped", "bordered", "condensed"),
      full_width = TRUE,
      font_size = 12) %>%
    add_header_above(header_top) %>%
    add_header_above(c(" " = 1, "Actual Class" = n_models * 2)) %>%
    column_spec(1, bold = TRUE) %>%
    row_spec(0, bold = TRUE, background = "#4472C4", color = "white")
)

# --- Performance Metrics Table ---
metrics_mat <- t(sapply(cms, get_metrics))

```

```

results_list[[cv]] <- metrics_mat

metrics_df      <- as.data.frame(metrics_mat)
metrics_df$Model <- rownames(metrics_df)
metrics_df      <- metrics_df[, c("Model","Accuracy","Precision","Recall","F1")]
rownames(metrics_df) <- NULL

best_f1_row <- which.max(metrics_df$F1)

cat("\n### Performance Metrics -", cv, "\n")
print(
  kable(metrics_df,
    caption = paste("Table: Q5 Performance Metrics -", cv),
    digits = 4,
    align = c("l","c","c","c","c")) %>%
  kable_styling(
    bootstrap_options = c("striped","hover","bordered","condensed"),
    full_width = FALSE,
    font_size = 13) %>%
  row_spec(0, bold = TRUE, background = "#4472C4", color = "white") %>%
  row_spec(best_f1_row, bold = TRUE, background = "#E8F5E9") %>%
  column_spec(1, bold = TRUE) %>%
  column_spec(5, bold = TRUE)
)
}

```

----- Class Variable: CWomensMvt -----

Confusion Matrices (All Models) - CWomensMvt

```

<table class="table table-striped table-bordered table-condensed" style="font-size: 12px;
margin-left: auto; margin-right: auto;">
<caption style="font-size: initial !important;">Confusion Matrices for all models -
CWomensMvt</caption>
  <thead>
<tr>
<th style="empty-cells: hide;border-bottom:hidden;" colspan="1"></th>
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-
align: center; " colspan="10"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;
">Actual Class</div></th>
</tr>
<tr>
<th style="empty-cells: hide;border-bottom:hidden;" colspan="1"></th>
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-
align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;
">Decision Tree</div></th>
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-
align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;
">Naive Bayes</div></th>
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-

```

```
align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;
">Bagging</div></th>
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-
align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;
">Boosting</div></th>
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-
align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;
">Random Forest</div></th>
</tr>
<tr>
<th style="text-align:left;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Predicted </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Low </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> High </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Low </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> High </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Low </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> High </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Low </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> High </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Low </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> High </th>
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<tbody>
<tr>
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<td style="text-align:center;"> 0 </td>
<td style="text-align:center;"> 0 </td>
<td style="text-align:center;"> 839 </td>
<td style="text-align:center;"> 909 </td>
<td style="text-align:center;"> 121 </td>
<td style="text-align:center;"> 136 </td>
<td style="text-align:center;"> 382 </td>
<td style="text-align:center;"> 322 </td>
<td style="text-align:center;"> 448 </td>
<td style="text-align:center;"> 393 </td>
</tr>
<tr>
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```

```

<td style="text-align:center;"> 1734 </td>
<td style="text-align:center;"> 2500 </td>
<td style="text-align:center;"> 895 </td>
<td style="text-align:center;"> 1591 </td>
<td style="text-align:center;"> 1613 </td>
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### Performance Metrics - CWomensMvt
<table class="table table-striped table-hover table-bordered table-condensed" style="font-size: 13px; width: auto !important; margin-left: auto; margin-right: auto;">
<caption style="font-size: initial !important;">Table: Q5 Performance Metrics - CWomensMvt</caption>
<thead>
<tr>
<th style="text-align:left;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> Model </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> Accuracy </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> Precision </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> Recall </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> F1 </th>
</tr>
</thead>
<tbody>
<tr>
<td style="text-align:left;font-weight: bold;background-color: rgba(232, 245, 233, 255) !important;font-weight: bold;"> Decision Tree </td>
<td style="text-align:center;font-weight: bold;background-color: rgba(232, 245, 233, 255) !important;"> 0.5905 </td>
<td style="text-align:center;font-weight: bold;background-color: rgba(232, 245, 233, 255) !important;"> 0.5905 </td>
<td style="text-align:center;font-weight: bold;background-color: rgba(232, 245, 233, 255) !important;"> 1.0000 </td>
<td style="text-align:center;font-weight: bold;background-color: rgba(232, 245, 233, 255) !important;font-weight: bold;"> 0.7425 </td>
</tr>
<tr>
<td style="text-align:left;font-weight: bold;"> Naive Bayes </td>
<td style="text-align:center;"> 0.5739 </td>
<td style="text-align:center;"> 0.6400 </td>
<td style="text-align:center;"> 0.6364 </td>

```

```

    <td style="text-align:center;font-weight: bold;"> 0.6382 </td>
</tr>
<tr>
    <td style="text-align:left;font-weight: bold;"> Bagging </td>
    <td style="text-align:center;"> 0.5869 </td>
    <td style="text-align:center;"> 0.5944 </td>
    <td style="text-align:center;"> 0.9456 </td>
    <td style="text-align:center;font-weight: bold;"> 0.7300 </td>
</tr>
<tr>
    <td style="text-align:left;font-weight: bold;"> Boosting </td>
    <td style="text-align:center;"> 0.6046 </td>
    <td style="text-align:center;"> 0.6170 </td>
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    <td style="text-align:center;font-weight: bold;"> 0.7224 </td>
</tr>
<tr>
    <td style="text-align:left;font-weight: bold;"> Random Forest </td>
    <td style="text-align:center;"> 0.6034 </td>
    <td style="text-align:center;"> 0.6210 </td>
    <td style="text-align:center;"> 0.8428 </td>
    <td style="text-align:center;font-weight: bold;"> 0.7151 </td>
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</tbody>
</table>
----- Class Variable: CGovernment -----

```

Confusion Matrices (All Models) - CGovernment

```

<table class="table table-striped table-bordered table-condensed" style="font-size: 12px;
margin-left: auto; margin-right: auto;">
<caption style="font-size: initial !important;">Confusion Matrices for all models -
CGovernment</caption>
  <thead>
<tr>
<th style="empty-cells: hide;border-bottom:hidden;" colspan="1"></th>
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-
align: center; " colspan="10"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;
">Actual Class</div></th>
</tr>
<tr>
<th style="empty-cells: hide;border-bottom:hidden;" colspan="1"></th>
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-
align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;
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<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-
align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;
">Naive Bayes</div></th>
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-
align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;
">Bagging</div></th>

```

```

--bg--" , "" , ""
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px; ">Boosting</div></th>
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px; ">Random Forest</div></th>
</tr>
<tr>
<th style="text-align:left;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> Predicted </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> Low </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> High </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> Low </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> High </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> Low </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> High </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> Low </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> High </th>
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<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> High </th>
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</thead>
<tbody>
<tr>
<td style="text-align:left;font-weight: bold;"> Low </td>
<td style="text-align:center;"> 1323 </td>
<td style="text-align:center;"> 868 </td>
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<tr>
<td style="text-align:left;font-weight: bold;"> High </td>
<td style="text-align:center;"> 974 </td>
<td style="text-align:center;"> 1060 </td>

```

```

<td style="text-align:center; / 100% /td>
<td style="text-align:center;"> 777 </td>
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<td style="text-align:center;"> 492 </td>
<td style="text-align:center;"> 671 </td>
<td style="text-align:center;"> 713 </td>
<td style="text-align:center;"> 935 </td>
<td style="text-align:center;"> 705 </td>
<td style="text-align:center;"> 922 </td>
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</table>
### Performance Metrics - CGovernment
<table class="table table-striped table-hover table-bordered table-condensed" style="font-
size: 13px; width: auto !important; margin-left: auto; margin-right: auto;">
<caption style="font-size: initial !important;">Table: Q5 Performance Metrics - CGovernment</
caption>
<thead>
<tr>
<th style="text-align:left;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Model </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Accuracy </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Precision </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Recall </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> F1 </th>
</tr>
</thead>
<tbody>
<tr>
<td style="text-align:left;font-weight: bold;background-color: rgba(232, 245, 233, 255) !
important;font-weight: bold;"> Decision Tree </td>
<td style="text-align:center;font-weight: bold;background-color: rgba(232, 245, 233, 255) !
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```

```

</tr>
<tr>
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<tr>
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  <td style="text-align:center;"> 0.5674 </td>
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  <td style="text-align:center;font-weight: bold;"> 0.5216 </td>
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<tr>
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  <td style="text-align:center;"> 0.5667 </td>
  <td style="text-align:center;"> 0.4760 </td>
  <td style="text-align:center;font-weight: bold;"> 0.5174 </td>
</tr>
</tbody>
</table>

```

----- Class Variable: CUnions -----

Confusion Matrices (All Models) - CUnions

```

<table class="table table-striped table-bordered table-condensed" style="font-size: 12px;
margin-left: auto; margin-right: auto;">
<caption style="font-size: initial !important;">Confusion Matrices for all models - CUnions</caption>
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<tr>
<th style="empty-cells: hide;border-bottom:hidden;" colspan="1"></th>
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;">Decision Tree</div></th>
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<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;">Bagging</div></th>
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;">Boosting</div></th>
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;">Random Forest</div></th>
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;">CUnions</div></th>
</tr>

```

```
align: center; " colspan="2"><div style="border-bottom: 1px solid #ddd; padding-bottom: 5px;
">Boosting</div></th>
<th style="border-bottom:hidden;padding-bottom:0; padding-left:3px;padding-right:3px;text-
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">Random Forest</div></th>
</tr>
<tr>
<th style="text-align:left;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Predicted </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Low </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> High </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Low </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
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<th style="text-align:center;font-weight: bold;color: white !important;background-color:
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<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> High </th>
</tr>
</thead>
<tbody>
<tr>
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<td style="text-align:center;"> 1232 </td>
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<tr>
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<td style="text-align:center;"> 0 </td>
<td style="text-align:center;"> 540 </td>
```

```
<td style="text-align:center;"> 498 </td>
<td style="text-align:center;"> 181 </td>
<td style="text-align:center;"> 236 </td>
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<td style="text-align:center;"> 432 </td>
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</tbody>
</table>
### Performance Metrics - CUnions
<table class="table table-striped table-hover table-bordered table-condensed" style="font-size: 13px; width: auto !important; margin-left: auto; margin-right: auto;">
<caption style="font-size: initial !important;">Table: Q5 Performance Metrics - CUnions</caption>
<thead>
<tr>
<th style="text-align:left;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> Model </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> Accuracy </th>
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<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> Recall </th>
<th style="text-align:center;font-weight: bold;color: white !important;background-color: rgba(68, 114, 196, 255) !important;"> F1 </th>
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<tbody>
<tr>
<td style="text-align:left;font-weight: bold;"> Decision Tree </td>
<td style="text-align:center;"> 0.5987 </td>
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<tr>
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<td style="text-align:center;"> 0.2931 </td>
<td style="text-align:center;font-weight: bold;"> 0.3639 </td>
</tr>
<tr>
<td style="text-align:left;font-weight: bold;"> Bagging </td>
<td style="text-align:center;"> 0.6117 </td>
<td style="text-align:center;"> 0.5659 </td>
<td style="text-align:center;"> 0.1389 </td>
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</tr>
```

```

</tr>
<tr>
  <td style="text-align:left;font-weight: bold;"> Boosting </td>
  <td style="text-align:center;"> 0.6039 </td>
  <td style="text-align:center;"> 0.5131 </td>
  <td style="text-align:center;"> 0.2543 </td>
  <td style="text-align:center;font-weight: bold;"> 0.3400 </td>
</tr>
<tr>
  <td style="text-align:left;font-weight: bold;background-color: rgba(232, 245, 233, 255) !
important;font-weight: bold;"> Random Forest </td>
  <td style="text-align:center;font-weight: bold;background-color: rgba(232, 245, 233, 255) !
important;"> 0.6169 </td>
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important;"> 0.2749 </td>
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important;font-weight: bold;"> 0.3654 </td>
</tr>
</tbody>
</table>

```

```

# =====
# QUESTION 6: ROC Curves & AUC
# =====
cat("\n===== QUESTION 6: ROC CURVES & AUC =====\n")

```

===== QUESTION 6: ROC CURVES & AUC =====

```

colors_roc <- c("blue", "red", "green3", "purple", "orange")
auc_list <- list()

# ---- Plot all 3 ROC curves first ----
for (cv in class_vars) {
  actual_bin <- ifelse(WD.test[[cv]] == "High", 1, 0)
  probs <- prob_list[[cv]]

  auc_vals <- numeric(length(model_names))
  names(auc_vals) <- model_names

  plot(0, 0, type = "n", xlim = c(0,1), ylim = c(0,1),
       xlab = "False Positive Rate", ylab = "True Positive Rate",
       main = paste("ROC Curves -", cv))
  abline(0, 1, lty = 2, col = "grey")

  for (i in seq_along(model_names)) {
    mn <- model_names[i]

```

```

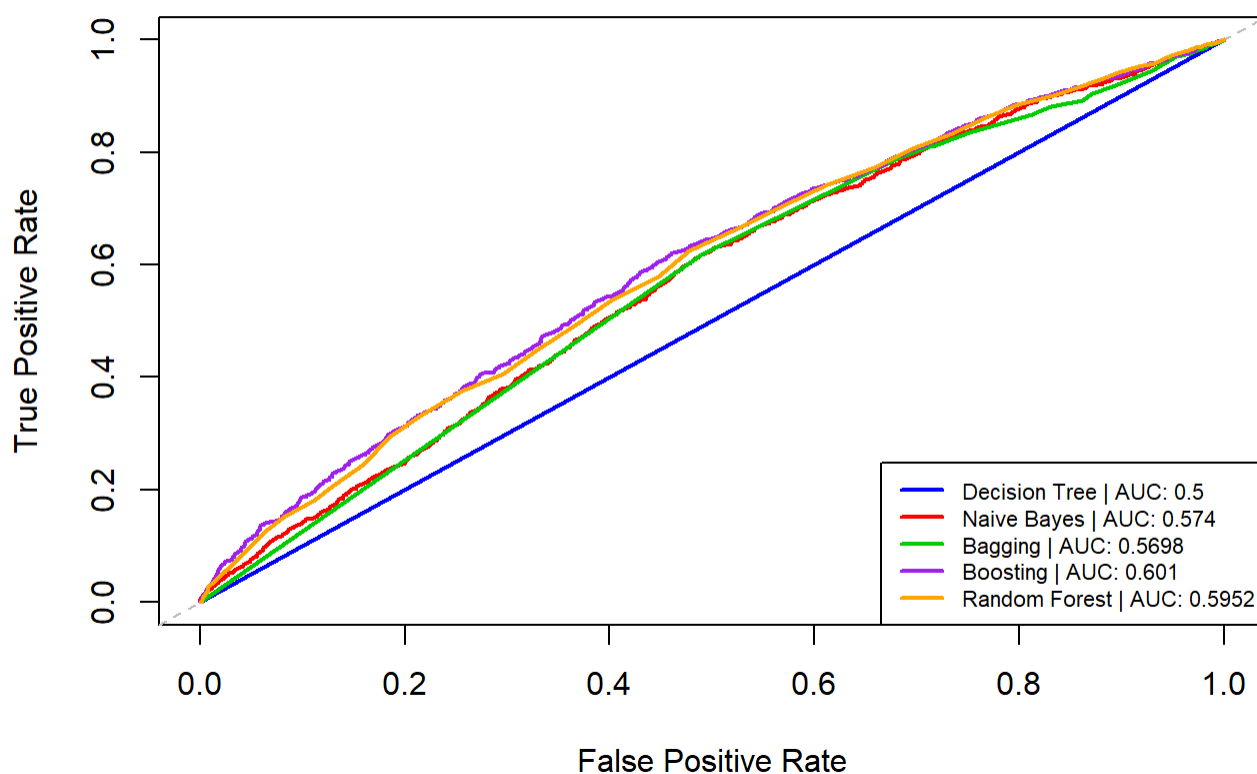
prd <- ROCR::prediction(probs[[mn]], actual_bin)
prf <- ROCR::performance(prd, "tpr", "fpr")
auc <- round(ROCR::performance(prd, "auc")@y.values[[1]], 4)
auc_vals[mn] <- auc
plot(prf, col = colors_roc[i], lwd = 2, add = TRUE)
}

legend("bottomright",
      legend = paste(model_names, "| AUC:", auc_vals),
      col     = colors_roc, lwd = 2, cex = 0.7)

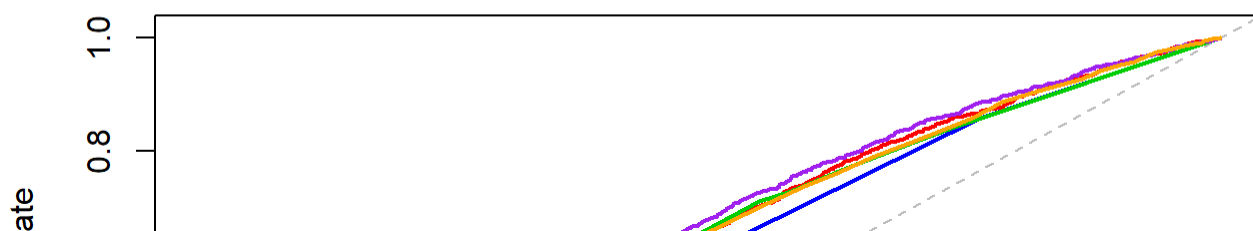
auc_list[[cv]] <- auc_vals
}

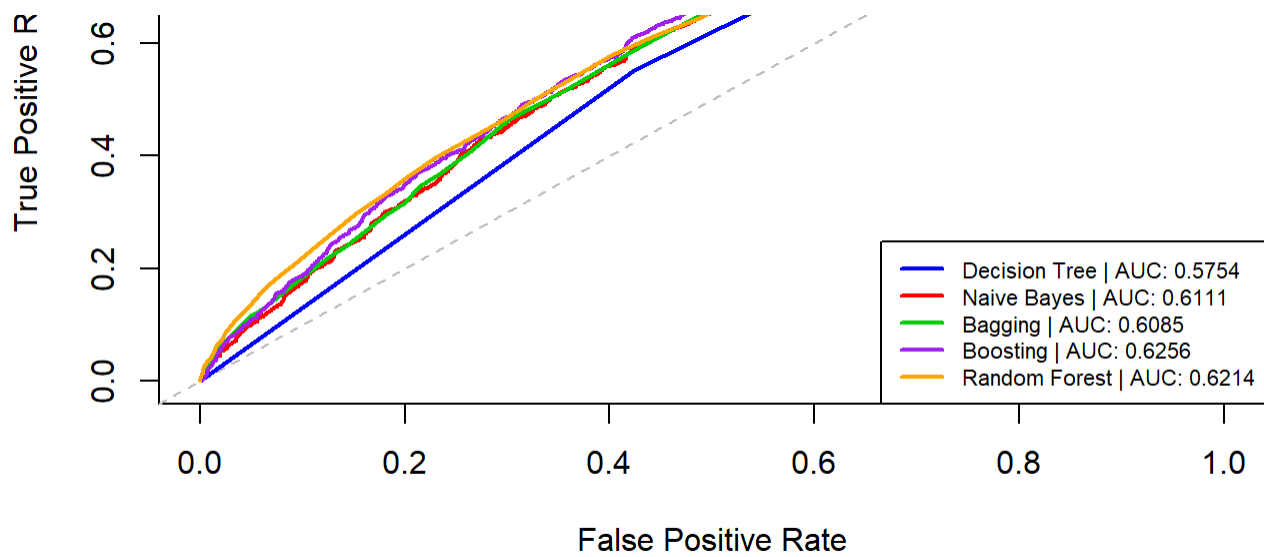
```

ROC Curves - CWomensMvt

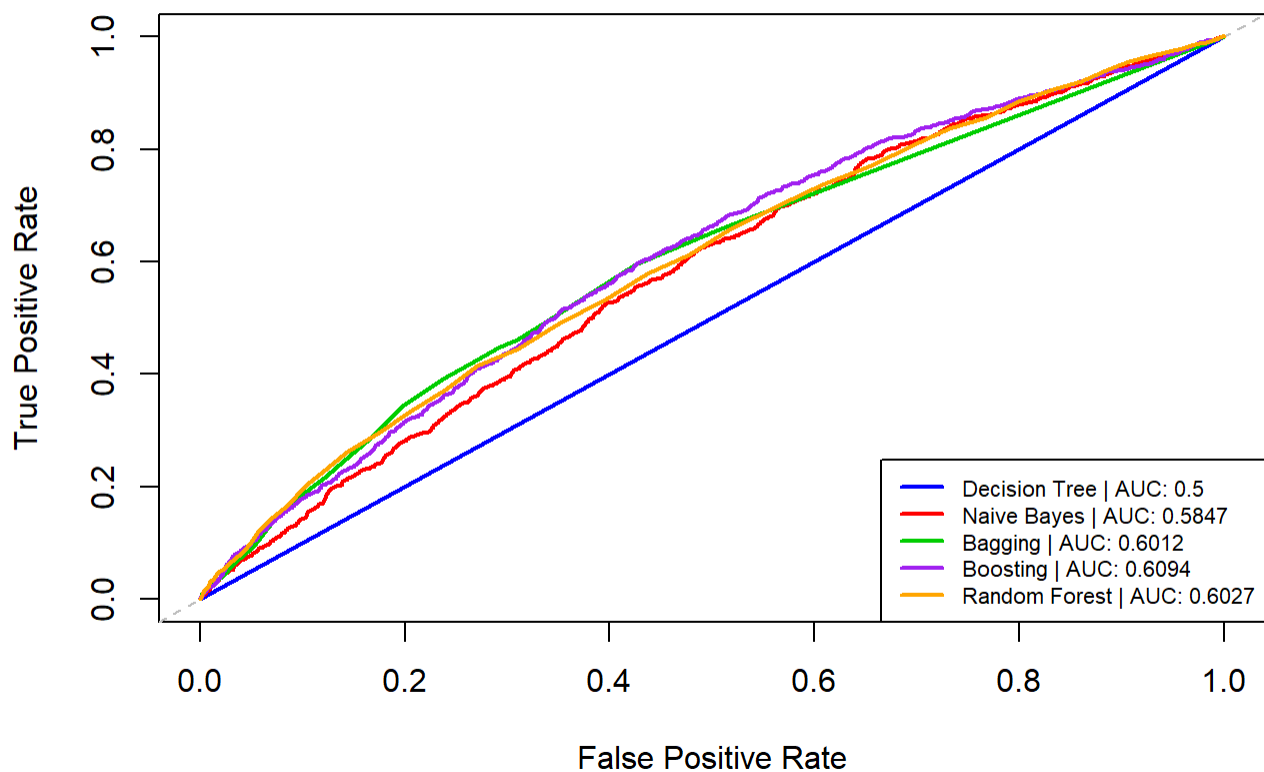


ROC Curves - CGovernment





ROC Curves - CUnions



```
# ---- Print all 3 AUC tables after ----
for (cv in class_vars) {
  cat("\nAUC values for", cv, ":\n")
  print(
    kable(data.frame(AUC = auc_list[[cv]]),
      caption = paste("Table: Q6 AUC -", cv),
      ...
    )
  )
}
```

```

        digits = 4) %>%
        kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                      full_width = FALSE)
    )
}

```

AUC values for CWomensMvt :

```

<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">

```

```

<caption>Table: Q6 AUC - CWomensMvt</caption>

```

```

<thead>

```

```

<tr>

```

```

  <th style="text-align:left;"> </th>

```

```

  <th style="text-align:right;"> AUC </th>

```

```

</tr>

```

```

</thead>

```

```

<tbody>

```

```

<tr>

```

```

  <td style="text-align:left;"> Decision Tree </td>

```

```

  <td style="text-align:right;"> 0.5000 </td>

```

```

</tr>

```

```

<tr>

```

```

  <td style="text-align:left;"> Naive Bayes </td>

```

```

  <td style="text-align:right;"> 0.5740 </td>

```

```

</tr>

```

```

<tr>

```

```

  <td style="text-align:left;"> Bagging </td>

```

```

  <td style="text-align:right;"> 0.5698 </td>

```

```

</tr>

```

```

<tr>

```

```

  <td style="text-align:left;"> Boosting </td>

```

```

  <td style="text-align:right;"> 0.6010 </td>

```

```

</tr>

```

```

<tr>

```

```

  <td style="text-align:left;"> Random Forest </td>

```

```

  <td style="text-align:right;"> 0.5952 </td>

```

```

</tr>

```

```

</tbody>

```

```

</table>

```

AUC values for CGovernment :

```

<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">

```

```

<caption>Table: Q6 AUC - CGovernment</caption>

```

```

<thead>

```

```

<tr>

```

```

  <th style="text-align:left;"> </th>

```

```

  <th style="text-align:right;"> AUC </th>

```

```

</tr>

```

```

</thead>

```

```

<tbody>
  <tr>
    <td style="text-align:left;"> Decision Tree </td>
    <td style="text-align:right;"> 0.5754 </td>
  </tr>
  <tr>
    <td style="text-align:left;"> Naive Bayes </td>
    <td style="text-align:right;"> 0.6111 </td>
  </tr>
  <tr>
    <td style="text-align:left;"> Bagging </td>
    <td style="text-align:right;"> 0.6085 </td>
  </tr>
  <tr>
    <td style="text-align:left;"> Boosting </td>
    <td style="text-align:right;"> 0.6256 </td>
  </tr>
  <tr>
    <td style="text-align:left;"> Random Forest </td>
    <td style="text-align:right;"> 0.6214 </td>
  </tr>
</tbody>
</table>

```

AUC values for CUnions :

```

<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">

```

```

<caption>Table: Q6 AUC - CUnions</caption>

```

```

<thead>
  <tr>
    <th style="text-align:left;"> </th>
    <th style="text-align:right;"> AUC </th>
  </tr>
</thead>
<tbody>
  <tr>
    <td style="text-align:left;"> Decision Tree </td>
    <td style="text-align:right;"> 0.5000 </td>
  </tr>
  <tr>
    <td style="text-align:left;"> Naive Bayes </td>
    <td style="text-align:right;"> 0.5847 </td>
  </tr>
  <tr>
    <td style="text-align:left;"> Bagging </td>
    <td style="text-align:right;"> 0.6012 </td>
  </tr>
  <tr>
    <td style="text-align:left;"> Boosting </td>
    <td style="text-align:right;"> 0.6094 </td>
  </tr>

```

```

<tr>
  <td style="text-align:left;"> Random Forest </td>
  <td style="text-align:right;"> 0.6027 </td>
</tr>
</tbody>
</table>

```

```

# =====
# QUESTION 7: Compare Classifiers
# =====
cat("\n===== QUESTION 7: CLASSIFIER COMPARISON =====\n")

```

===== QUESTION 7: CLASSIFIER COMPARISON =====

```

# Full summary table per class variable (metrics + AUC)
for (cv in class_vars) {
  cat("\n--- Full performance summary for:", cv, "---\n")
  df      <- as.data.frame(results_list[[cv]])
  df$AUC  <- auc_list[[cv]]
  print(kable(df,
              caption = paste("Table: Q7 Full Metrics -", cv),
              digits  = 4) %>%
        kable_styling(bootstrap_options = c("striped","hover","condensed"),
                      full_width = FALSE))
}

```

```

--- Full performance summary for: CWomensMvt ---
<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">
<caption>Table: Q7 Full Metrics - CWomensMvt</caption>
<thead>
<tr>
  <th style="text-align:left;"> </th>
  <th style="text-align:right;"> Accuracy </th>
  <th style="text-align:right;"> Precision </th>
  <th style="text-align:right;"> Recall </th>
  <th style="text-align:right;"> F1 </th>
  <th style="text-align:right;"> AUC </th>
</tr>
</thead>
<tbody>
<tr>
  <td style="text-align:left;"> Decision Tree </td>
  <td style="text-align:right;"> 0.5905 </td>
  <td style="text-align:right;"> 0.5905 </td>
  <td style="text-align:right;"> 1.0000 </td>
  <td style="text-align:right;"> 0.7425 </td>

```

```
 0.5000 |

|               |        |        |        |        |        |
|---------------|--------|--------|--------|--------|--------|
| Naive Bayes   | 0.5739 | 0.6400 | 0.6364 | 0.6382 | 0.5740 |
| Bagging       | 0.5869 | 0.5944 | 0.9456 | 0.7300 | 0.5698 |
| Boosting      | 0.6046 | 0.6170 | 0.8712 | 0.7224 | 0.6010 |
| Random Forest | 0.6034 | 0.6210 | 0.8428 | 0.7151 | 0.5952 |


```

```


```

--- Full performance summary for: CGovernment ---

Table: Q7 Full Metrics - CGovernment					
	Accuracy	Precision	Recall	F1	AUC

```


```

```

</thead>
<tbody>
  <tr>
    <td style="text-align:left;"> Decision Tree </td>
    <td style="text-align:right;"> 0.5650 </td>
    <td style="text-align:right;"> 0.5233 </td>
    <td style="text-align:right;"> 0.5519 </td>
    <td style="text-align:right;"> 0.5372 </td>
    <td style="text-align:right;"> 0.5754 </td>
  </tr>
  <tr>
    <td style="text-align:left;"> Naive Bayes </td>
    <td style="text-align:right;"> 0.5864 </td>
    <td style="text-align:right;"> 0.5534 </td>
    <td style="text-align:right;"> 0.4972 </td>
    <td style="text-align:right;"> 0.5238 </td>
    <td style="text-align:right;"> 0.6111 </td>
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    <td style="text-align:right;"> 0.5770 </td>
    <td style="text-align:right;"> 0.3464 </td>
    <td style="text-align:right;"> 0.4329 </td>
    <td style="text-align:right;"> 0.6085 </td>
  </tr>
  <tr>
    <td style="text-align:left;"> Boosting </td>
    <td style="text-align:right;"> 0.5949 </td>
    <td style="text-align:right;"> 0.5674 </td>
    <td style="text-align:right;"> 0.4827 </td>
    <td style="text-align:right;"> 0.5216 </td>
    <td style="text-align:right;"> 0.6256 </td>
  </tr>
  <tr>
    <td style="text-align:left;"> Random Forest </td>
    <td style="text-align:right;"> 0.5938 </td>
    <td style="text-align:right;"> 0.5667 </td>
    <td style="text-align:right;"> 0.4760 </td>
    <td style="text-align:right;"> 0.5174 </td>
    <td style="text-align:right;"> 0.6214 </td>
  </tr>
</tbody>
</table>
--- Full performance summary for: CUnions ---
<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">
<caption>Table: Q7 Full Metrics - CUnions</caption>
<thead>
  <tr>
    <th>Model</th>
    <th>Accuracy</th>
    <th>Precision</th>
    <th>Recall</th>
    <th>F1 Score</th>
    <th>ROC AUC</th>
  </tr>

```

```
<tn style="text-align:left;"> </tn>
<th style="text-align:right;"> Accuracy </th>
<th style="text-align:right;"> Precision </th>
<th style="text-align:right;"> Recall </th>
<th style="text-align:right;"> F1 </th>
<th style="text-align:right;"> AUC </th>
</tr>
</thead>
<tbody>
<tr>
<td style="text-align:left;"> Decision Tree </td>
<td style="text-align:right;"> 0.5987 </td>
<td style="text-align:right;"> 0.0000 </td>
<td style="text-align:right;"> 0.0000 </td>
<td style="text-align:right;"> 0.0000 </td>
<td style="text-align:right;"> 0.5000 </td>
</tr>
<tr>
<td style="text-align:left;"> Naive Bayes </td>
<td style="text-align:right;"> 0.5888 </td>
<td style="text-align:right;"> 0.4798 </td>
<td style="text-align:right;"> 0.2931 </td>
<td style="text-align:right;"> 0.3639 </td>
<td style="text-align:right;"> 0.5847 </td>
</tr>
<tr>
<td style="text-align:left;"> Bagging </td>
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<td style="text-align:right;"> 0.5659 </td>
<td style="text-align:right;"> 0.1389 </td>
<td style="text-align:right;"> 0.2231 </td>
<td style="text-align:right;"> 0.6012 </td>
</tr>
<tr>
<td style="text-align:left;"> Boosting </td>
<td style="text-align:right;"> 0.6039 </td>
<td style="text-align:right;"> 0.5131 </td>
<td style="text-align:right;"> 0.2543 </td>
<td style="text-align:right;"> 0.3400 </td>
<td style="text-align:right;"> 0.6094 </td>
</tr>
<tr>
<td style="text-align:left;"> Random Forest </td>
<td style="text-align:right;"> 0.6169 </td>
<td style="text-align:right;"> 0.5449 </td>
<td style="text-align:right;"> 0.2749 </td>
<td style="text-align:right;"> 0.3654 </td>
<td style="text-align:right;"> 0.6027 </td>
</tr>
</tbody>
```

</table>

```
# Best & worst classifier per class variable by F1
cat("\n--- Best & Worst Classifiers by F1-score ---\n")
```

```
--- Best & Worst Classifiers by F1-score ---
```

```
bw_df <- data.frame()
for (cv in class_vars) {
  f1s <- results_list[[cv]][, "F1"]
  best <- names(which.max(f1s))
  worst <- names(which.min(f1s))
  bw_df <- rbind(bw_df,
                 data.frame(ClassVar = cv,
                           Best      = best, BestF1 = round(max(f1s), 4),
                           Worst     = worst, WorstF1 = round(min(f1s), 4)))
}
print(kable(bw_df,
            caption = "Table: Q7 Best and Worst Classifiers by F1") %>%
      kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                    full_width = FALSE))
```

```
<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">
```

```
<caption>Table: Q7 Best and Worst Classifiers by F1</caption>
```

```
<thead>
```

```
<tr>
```

```
<th style="text-align:left;"> ClassVar </th>
```

```
<th style="text-align:left;"> Best </th>
```

```
<th style="text-align:right;"> BestF1 </th>
```

```
<th style="text-align:left;"> Worst </th>
```

```
<th style="text-align:right;"> WorstF1 </th>
```

```
</tr>
```

```
</thead>
```

```
<tbody>
```

```
<tr>
```

```
<td style="text-align:left;"> CWomensMvt </td>
```

```
<td style="text-align:left;"> Decision Tree </td>
```

```
<td style="text-align:right;"> 0.7425 </td>
```

```
<td style="text-align:left;"> Naive Bayes </td>
```

```
<td style="text-align:right;"> 0.6382 </td>
```

```
</tr>
```

```
<tr>
```

```
<td style="text-align:left;"> CGovernment </td>
```

```
<td style="text-align:left;"> Decision Tree </td>
```

```
<td style="text-align:right;"> 0.5372 </td>
```

```
<td style="text-align:left;"> Bagging </td>
```

```
<td style="text-align:right;"> 0.4329 </td>
```

```

</tr>
<tr>
  <td style="text-align:left;"> CUnions </td>
  <td style="text-align:left;"> Random Forest </td>
  <td style="text-align:right;"> 0.3654 </td>
  <td style="text-align:left;"> Decision Tree </td>
  <td style="text-align:right;"> 0.0000 </td>
</tr>
</tbody>
</table>

```

```

# Worst ensemble (Bagging / Boosting / Random Forest) by average F1
cat("\n--- Worst Ensemble by Average F1 across all class variables ---\n")

```

--- Worst Ensemble by Average F1 across all class variables ---

```

ens_names <- c("Bagging", "Boosting", "Random Forest")
ens_avg   <- sapply(ens_names, function(en) {
  mean(sapply(class_vars, function(cv) results_list[[cv]][en, "F1"]))
})
print(kable(data.frame(Ensemble = ens_names, Avg_F1 = round(ens_avg, 4)),
  caption = "Table: Q7 Average F1 per Ensemble Model") %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
    full_width = FALSE))

```

```

<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">
<caption>Table: Q7 Average F1 per Ensemble Model</caption>
<thead>
  <tr>
    <th style="text-align:left;"> </th>
    <th style="text-align:left;"> Ensemble </th>
    <th style="text-align:right;"> Avg_F1 </th>
  </tr>
</thead>
<tbody>
  <tr>
    <td style="text-align:left;"> Bagging </td>
    <td style="text-align:left;"> Bagging </td>
    <td style="text-align:right;"> 0.4620 </td>
  </tr>
  <tr>
    <td style="text-align:left;"> Boosting </td>
    <td style="text-align:left;"> Boosting </td>
    <td style="text-align:right;"> 0.5280 </td>
  </tr>
  <tr>
    <td style="text-align:left;"> Random Forest </td>

```

```

    <td style="text-align:left;"> Random Forest </td>
    <td style="text-align:right;"> 0.5326 </td>
  </tr>
</tbody>
</table>

```

```

worst_ens <- names(which.min(ens_avg))
cat("Worst ensemble overall:", worst_ens, "\n")

```

Worst ensemble overall: Bagging

```

# =====
# QUESTION 8: Feature Importance (Ensemble Models)
# =====
cat("\n===== QUESTION 8: FEATURE IMPORTANCE =====\n")

```

===== QUESTION 8: FEATURE IMPORTANCE =====

```

for (cv in class_vars) {
  cat("\n----- Feature Importance for:", cv, "-----\n")

  bag_obj <- get(paste0("bag_", cv))
  bst_obj <- get(paste0("bst_", cv))
  rf_obj  <- get(paste0("rf_",  cv))

  # adabag stores $importance as a named numeric vector
  bag_imp <- sort(bag_obj$importance, decreasing = TRUE)
  bst_imp <- sort(bst_obj$importance, decreasing = TRUE)
  rf_imp  <- sort(importance(rf_obj)[, "MeanDecreaseGini"], decreasing = TRUE)

  bag_df <- data.frame(Attribute = names(bag_imp),
                      Importance = round(as.vector(bag_imp), 4))
  bst_df <- data.frame(Attribute = names(bst_imp),
                      Importance = round(as.vector(bst_imp), 4))
  rf_df  <- data.frame(Attribute = names(rf_imp),
                      Importance = round(rf_imp, 4))

  # Top-10 combined table
  imp_combined <- data.frame(
    Rank      = 1:10,
    Bagging   = head(bag_df$Attribute, 10),
    Boosting  = head(bst_df$Attribute, 10),
    RF        = head(rf_df$Attribute, 10)
  )
  print(kable(imp_combined,
              caption = paste("Table: Q8 Top-10 Important Attributes -", cv)) %>%
        kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                      full_width = FALSE))
}

```

```

full_width = FALSE,,

# Bar charts for all three ensemble models side by side
par(mfrow = c(1, 3), mar = c(8, 4, 3, 1))

barplot(head(bag_df$Importance, 15),
        names.arg = head(bag_df$Attribute, 15),
        las = 2, col = "steelblue", cex.names = 0.65,
        main = paste("(a) Bagging -", cv),
        ylab = "Importance")

barplot(head(bst_df$Importance, 15),
        names.arg = head(bst_df$Attribute, 15),
        las = 2, col = "tomato", cex.names = 0.65,
        main = paste("(b) Boosting -", cv),
        ylab = "Importance")

barplot(head(rf_df$Importance, 15),
        names.arg = head(rf_df$Attribute, 15),
        las = 2, col = "seagreen", cex.names = 0.65,
        main = paste("(c) Random Forest -", cv),
        ylab = "Mean Decrease Gini")

mtext(paste("Figure: Feature Importance -", cv),
      outer = TRUE, line = -1.5, cex = 0.9)
par(mfrow = c(1, 1), mar = c(5, 4, 4, 2))
}

```

----- Feature Importance for: CWomensMvt -----

<table class="table table-striped table-hover table-condensed" style="width: auto !important; margin-left: auto; margin-right: auto;">

<caption>Table: Q8 Top-10 Important Attributes - CWomensMvt</caption>

<thead>

<tr>

<th style="text-align:right;"> Rank </th>

<th style="text-align:left;"> Bagging </th>

<th style="text-align:left;"> Boosting </th>

<th style="text-align:left;"> RF </th>

</tr>

</thead>

<tbody>

<tr>

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<td style="text-align:left;"> PolDemoc </td>

<td style="text-align:left;"> PolDemoc </td>

<td style="text-align:left;"> Age </td>

</tr>

<tr>

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```

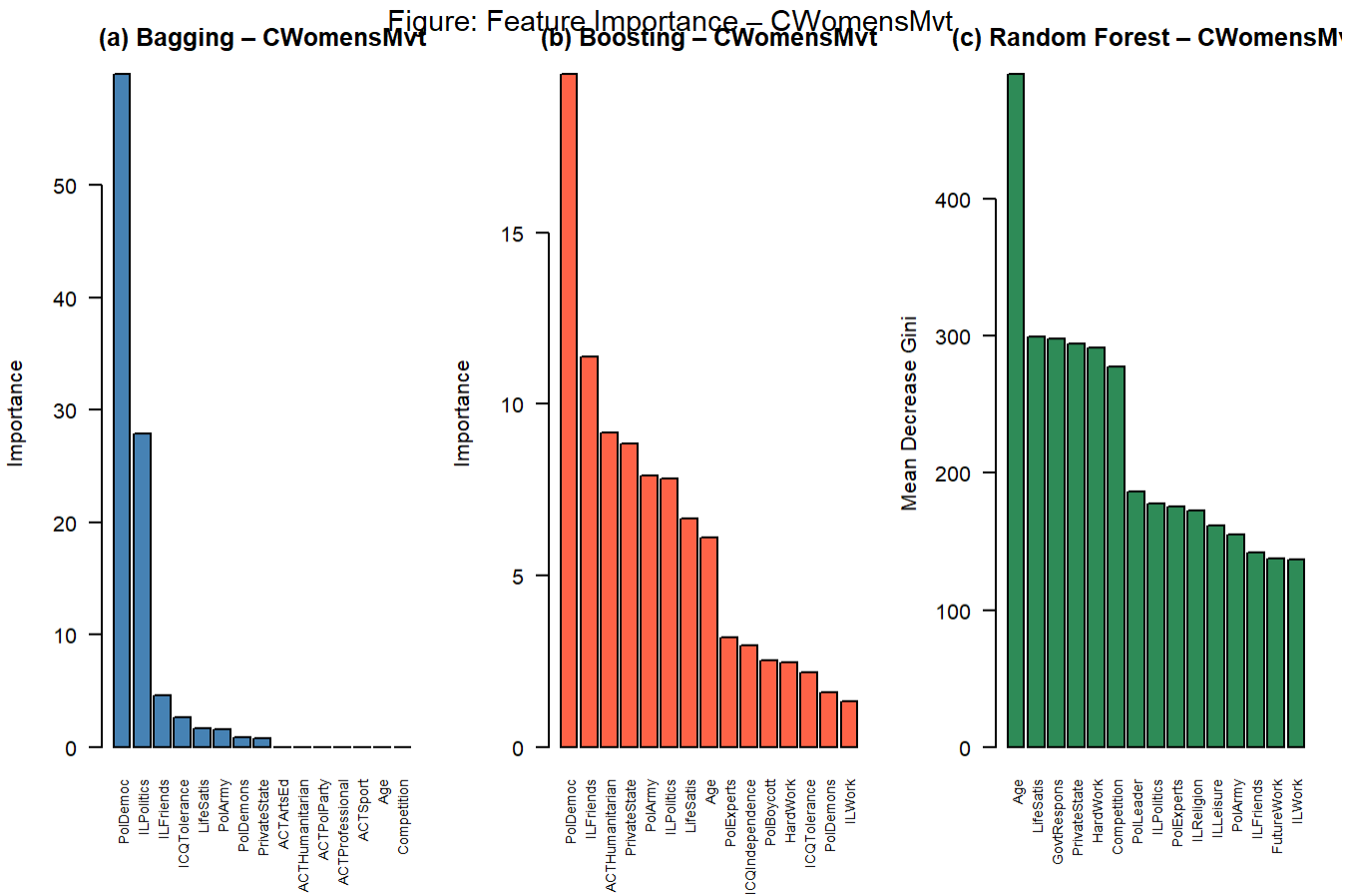
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<td style="text-align:left;"> ILFriends </td>
<td style="text-align:left;"> LifeSatis </td>
</tr>
<tr>
<td style="text-align:right;"> 3 </td>
<td style="text-align:left;"> ILFriends </td>
<td style="text-align:left;"> ACTHumanitarian </td>
<td style="text-align:left;"> GovtRespons </td>
</tr>
<tr>
<td style="text-align:right;"> 4 </td>
<td style="text-align:left;"> ICQTolerance </td>
<td style="text-align:left;"> PrivateState </td>
<td style="text-align:left;"> PrivateState </td>
</tr>
<tr>
<td style="text-align:right;"> 5 </td>
<td style="text-align:left;"> LifeSatis </td>
<td style="text-align:left;"> PolArmy </td>
<td style="text-align:left;"> HardWork </td>
</tr>
<tr>
<td style="text-align:right;"> 6 </td>
<td style="text-align:left;"> PolArmy </td>
<td style="text-align:left;"> ILPolitics </td>
<td style="text-align:left;"> Competition </td>
</tr>
<tr>
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<td style="text-align:left;"> LifeSatis </td>
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</tr>
<tr>
<td style="text-align:right;"> 8 </td>
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<tr>
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<td style="text-align:left;"> ICQIndependence </td>

```

```

    <td style="text-align:left;"> ILReligion </td>
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</table>

```



----- Feature Importance for: CGovernment -----

```

<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">

```

```

<caption>Table: Q8 Top-10 Important Attributes – CGovernment</caption>

```

```

<thead>

```

```

  <tr>

```

```

    <th style="text-align:right;"> Rank </th>

```

```

    <th style="text-align:left;"> Bagging </th>

```

```

    <th style="text-align:left;"> Boosting </th>

```

```

    <th style="text-align:left;"> RF </th>

```

```

  </tr>

```

```

</thead>

```

```

<tbody>

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```

```

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```

```

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```

```

    <td style="text-align:left;"> ILPolitics </td>

```

```

    <td style="text-align:left;"> Age </td>

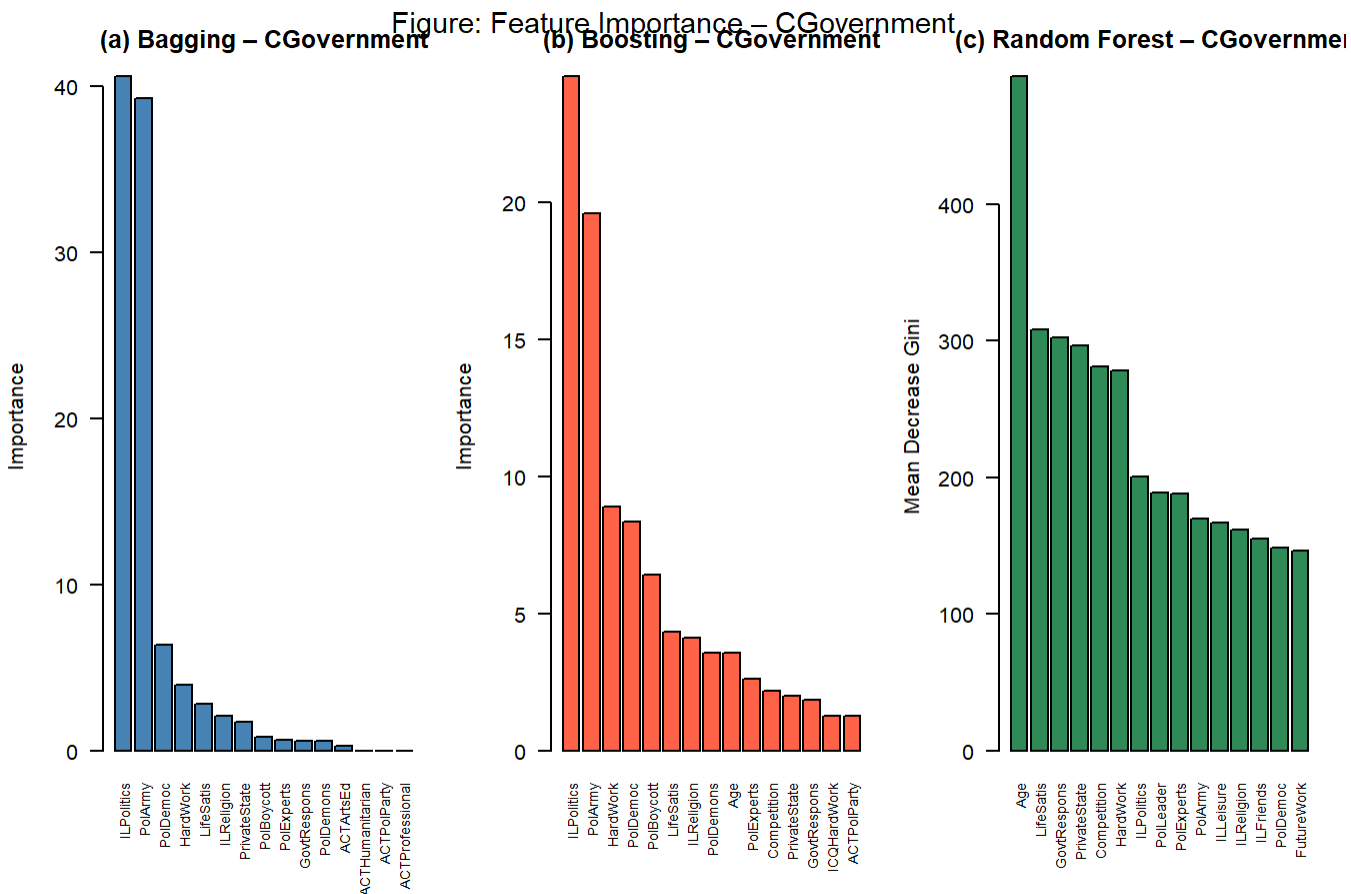
```

```
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<tr>
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  <td style="text-align:left;"> PolArmy </td>
  <td style="text-align:left;"> LifeSatis </td>
</tr>
<tr>
  <td style="text-align:right;"> 3 </td>
  <td style="text-align:left;"> PolDemoc </td>
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</tr>
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<tr>
  <td style="text-align:right;"> 5 </td>
  <td style="text-align:left;"> LifeSatis </td>
  <td style="text-align:left;"> PolBoycott </td>
  <td style="text-align:left;"> Competition </td>
</tr>
<tr>
  <td style="text-align:right;"> 6 </td>
  <td style="text-align:left;"> ILReligion </td>
  <td style="text-align:left;"> LifeSatis </td>
  <td style="text-align:left;"> HardWork </td>
</tr>
<tr>
  <td style="text-align:right;"> 7 </td>
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  <td style="text-align:left;"> ILReligion </td>
  <td style="text-align:left;"> ILPolitics </td>
</tr>
<tr>
  <td style="text-align:right;"> 8 </td>
  <td style="text-align:left;"> PolBoycott </td>
  <td style="text-align:left;"> PolDemons </td>
  <td style="text-align:left;"> PolLeader </td>
</tr>
<tr>
  <td style="text-align:right;"> 9 </td>
  <td style="text-align:left;"> PolExperts </td>
  <td style="text-align:left;"> Age </td>
  <td style="text-align:left;"> PolExperts </td>
</tr>
<tr>
```

```

<td style="text-align:right;"> 10 </td>
<td style="text-align:left;"> GovtRespons </td>
<td style="text-align:left;"> PolExperts </td>
<td style="text-align:left;"> PolArmy </td>
</tr>
</tbody>
</table>

```



----- Feature Importance for: CUnions -----

```

<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">

```

<caption>Table: Q8 Top-10 Important Attributes – CUnions</caption>

```

<thead>

```

```

<tr>

```

```

<th style="text-align:right;"> Rank </th>

```

```

<th style="text-align:left;"> Bagging </th>

```

```

<th style="text-align:left;"> Boosting </th>

```

```

<th style="text-align:left;"> RF </th>

```

```

</tr>

```

```

</thead>

```

```

<tbody>

```

```

<tr>

```

```

<td style="text-align:right;"> 1 </td>

```

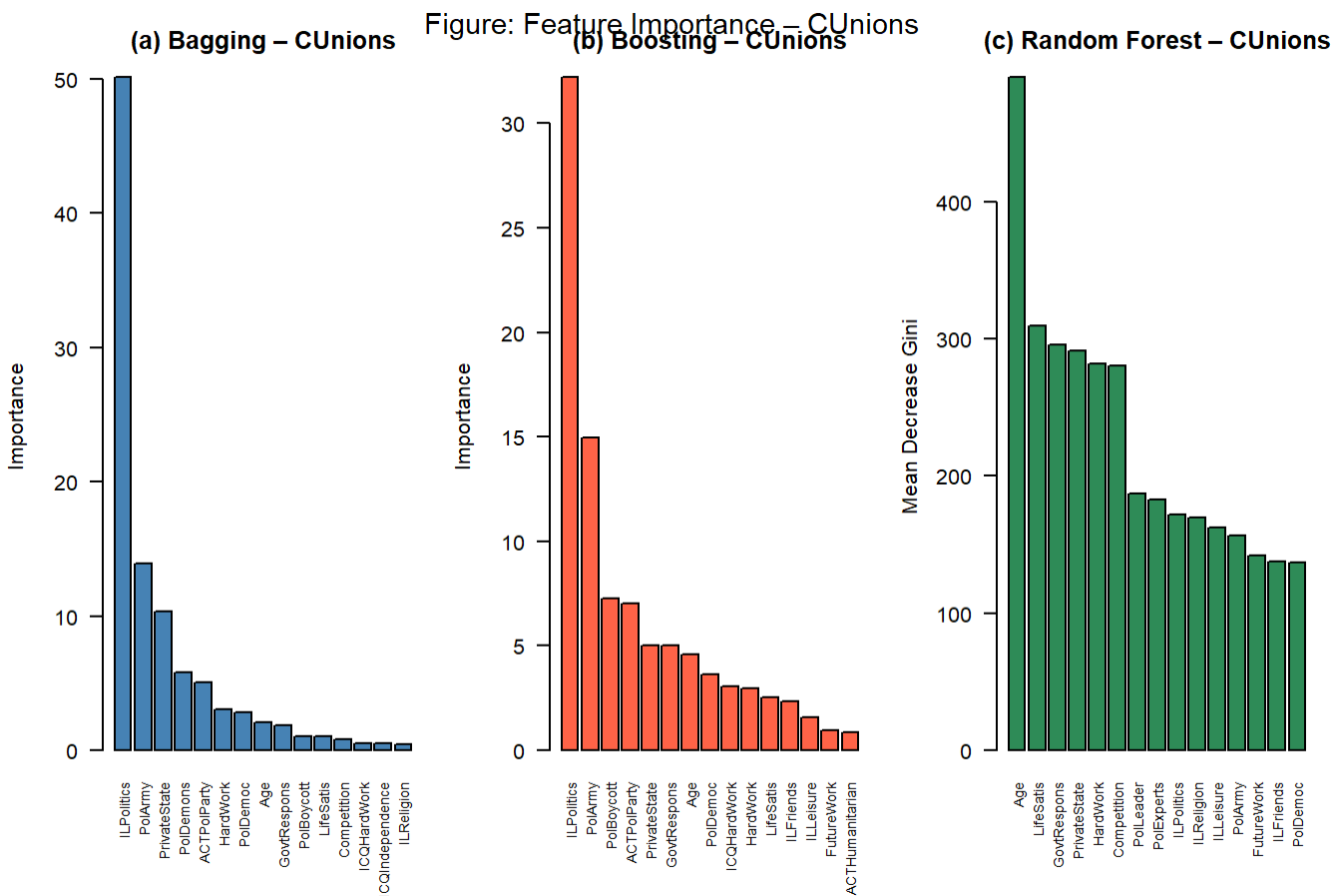
```
 1 | ILPolitics | ILPolitics | Age || 2 | PolArmy | PolArmy | LifeSatis |
| 3 | PrivateState | PolBoycott | GovtRespons |
| 4 | PolDemons | ACTPolParty | PrivateState |
| 5 | ACTPolParty | PrivateState | HardWork |
| 6 | HardWork | GovtRespons | Competition |
| 7 | PolDemoc | Age | PolLeader |
| 8 | Age | PolDemoc | PolExperts |
| 9 | GovtRespons | TCOHeadline |  |

```

```

<td style="text-align:left;"> ICQHardWork </td>
<td style="text-align:left;"> ILPolitics </td>
</tr>
<tr>
<td style="text-align:right;"> 10 </td>
<td style="text-align:left;"> PolBoycott </td>
<td style="text-align:left;"> HardWork </td>
<td style="text-align:left;"> ILReligion </td>
</tr>
</tbody>
</table>

```



```

# =====
# QUESTION 9: Strengths & Weaknesses Reference Tables
# =====
cat("\n===== QUESTION 9: MODEL COMPARISON SUMMARY =====\n")

```

```
===== QUESTION 9: MODEL COMPARISON SUMMARY =====
```

```

for (cv in class_vars) {
  cat("\n--- Full performance summary for:", cv, "---\n")
  df      <- as.data.frame(results list[[cv]])
}

```

```

df$AUC <- auc_list[[cv]]
print(kable(df,
            caption = paste("Table: Q9 Reference -", cv),
            digits = 4) %>%
      kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                    full_width = FALSE))
}

```

--- Full performance summary for: CWomensMvt ---

```

<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">

```

```

<caption>Table: Q9 Reference - CWomensMvt</caption>

```

```

<thead>

```

```

<tr>

```

```

  <th style="text-align:left;"> </th>

```

```

  <th style="text-align:right;"> Accuracy </th>

```

```

  <th style="text-align:right;"> Precision </th>

```

```

  <th style="text-align:right;"> Recall </th>

```

```

  <th style="text-align:right;"> F1 </th>

```

```

  <th style="text-align:right;"> AUC </th>

```

```

</tr>

```

```

</thead>

```

```

<tbody>

```

```

<tr>

```

```

  <td style="text-align:left;"> Decision Tree </td>

```

```

  <td style="text-align:right;"> 0.5905 </td>

```

```

  <td style="text-align:right;"> 0.5905 </td>

```

```

  <td style="text-align:right;"> 1.0000 </td>

```

```

  <td style="text-align:right;"> 0.7425 </td>

```

```

  <td style="text-align:right;"> 0.5000 </td>

```

```

</tr>

```

```

<tr>

```

```

  <td style="text-align:left;"> Naive Bayes </td>

```

```

  <td style="text-align:right;"> 0.5739 </td>

```

```

  <td style="text-align:right;"> 0.6400 </td>

```

```

  <td style="text-align:right;"> 0.6364 </td>

```

```

  <td style="text-align:right;"> 0.6382 </td>

```

```

  <td style="text-align:right;"> 0.5740 </td>

```

```

</tr>

```

```

<tr>

```

```

  <td style="text-align:left;"> Bagging </td>

```

```

  <td style="text-align:right;"> 0.5869 </td>

```

```

  <td style="text-align:right;"> 0.5944 </td>

```

```

  <td style="text-align:right;"> 0.9456 </td>

```

```

  <td style="text-align:right;"> 0.7300 </td>

```

```

  <td style="text-align:right;"> 0.5698 </td>

```

```

</tr>

```

```

<tr>

```

```

  <td style="text-align:left;"> </td>

```

```

  <td style="text-align:right;"> </td>

```

```

  <td style="text-align:right;"> </td>

```

```

<td style="text-align:left;"> Boosting </td>
<td style="text-align:right;"> 0.6046 </td>
<td style="text-align:right;"> 0.6170 </td>
<td style="text-align:right;"> 0.8712 </td>
<td style="text-align:right;"> 0.7224 </td>
<td style="text-align:right;"> 0.6010 </td>
</tr>
<tr>
<td style="text-align:left;"> Random Forest </td>
<td style="text-align:right;"> 0.6034 </td>
<td style="text-align:right;"> 0.6210 </td>
<td style="text-align:right;"> 0.8428 </td>
<td style="text-align:right;"> 0.7151 </td>
<td style="text-align:right;"> 0.5952 </td>
</tr>
</tbody>
</table>
--- Full performance summary for: CGovernment ---
<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">
<caption>Table: Q9 Reference – CGovernment</caption>
<thead>
<tr>
<th style="text-align:left;"> </th>
<th style="text-align:right;"> Accuracy </th>
<th style="text-align:right;"> Precision </th>
<th style="text-align:right;"> Recall </th>
<th style="text-align:right;"> F1 </th>
<th style="text-align:right;"> AUC </th>
</tr>
</thead>
<tbody>
<tr>
<td style="text-align:left;"> Decision Tree </td>
<td style="text-align:right;"> 0.5650 </td>
<td style="text-align:right;"> 0.5233 </td>
<td style="text-align:right;"> 0.5519 </td>
<td style="text-align:right;"> 0.5372 </td>
<td style="text-align:right;"> 0.5754 </td>
</tr>
<tr>
<td style="text-align:left;"> Naive Bayes </td>
<td style="text-align:right;"> 0.5864 </td>
<td style="text-align:right;"> 0.5534 </td>
<td style="text-align:right;"> 0.4972 </td>
<td style="text-align:right;"> 0.5238 </td>
<td style="text-align:right;"> 0.6111 </td>
</tr>
<tr>
<td style="text-align:left;"> Bagging </td>

```

```
<td style="text-align:right;"> 0.5848 </td>
<td style="text-align:right;"> 0.5770 </td>
<td style="text-align:right;"> 0.3464 </td>
<td style="text-align:right;"> 0.4329 </td>
<td style="text-align:right;"> 0.6085 </td>
</tr>
<tr>
<td style="text-align:left;"> Boosting </td>
<td style="text-align:right;"> 0.5949 </td>
<td style="text-align:right;"> 0.5674 </td>
<td style="text-align:right;"> 0.4827 </td>
<td style="text-align:right;"> 0.5216 </td>
<td style="text-align:right;"> 0.6256 </td>
</tr>
<tr>
<td style="text-align:left;"> Random Forest </td>
<td style="text-align:right;"> 0.5938 </td>
<td style="text-align:right;"> 0.5667 </td>
<td style="text-align:right;"> 0.4760 </td>
<td style="text-align:right;"> 0.5174 </td>
<td style="text-align:right;"> 0.6214 </td>
</tr>
</tbody>
</table>
--- Full performance summary for: CUnions ---
<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">
<caption>Table: Q9 Reference - CUnions</caption>
<thead>
<tr>
<th style="text-align:left;"> </th>
<th style="text-align:right;"> Accuracy </th>
<th style="text-align:right;"> Precision </th>
<th style="text-align:right;"> Recall </th>
<th style="text-align:right;"> F1 </th>
<th style="text-align:right;"> AUC </th>
</tr>
</thead>
<tbody>
<tr>
<td style="text-align:left;"> Decision Tree </td>
<td style="text-align:right;"> 0.5987 </td>
<td style="text-align:right;"> 0.0000 </td>
<td style="text-align:right;"> 0.0000 </td>
<td style="text-align:right;"> 0.0000 </td>
<td style="text-align:right;"> 0.5000 </td>
</tr>
<tr>
<td style="text-align:left;"> Naive Bayes </td>
<td style="text-align:right;"> 0.5888 </td>
```

```

<td style="text-align:right;"> 0.4798 </td>
<td style="text-align:right;"> 0.2931 </td>
<td style="text-align:right;"> 0.3639 </td>
<td style="text-align:right;"> 0.5847 </td>
</tr>
<tr>
<td style="text-align:left;"> Bagging </td>
<td style="text-align:right;"> 0.6117 </td>
<td style="text-align:right;"> 0.5659 </td>
<td style="text-align:right;"> 0.1389 </td>
<td style="text-align:right;"> 0.2231 </td>
<td style="text-align:right;"> 0.6012 </td>
</tr>
<tr>
<td style="text-align:left;"> Boosting </td>
<td style="text-align:right;"> 0.6039 </td>
<td style="text-align:right;"> 0.5131 </td>
<td style="text-align:right;"> 0.2543 </td>
<td style="text-align:right;"> 0.3400 </td>
<td style="text-align:right;"> 0.6094 </td>
</tr>
<tr>
<td style="text-align:left;"> Random Forest </td>
<td style="text-align:right;"> 0.6169 </td>
<td style="text-align:right;"> 0.5449 </td>
<td style="text-align:right;"> 0.2749 </td>
<td style="text-align:right;"> 0.3654 </td>
<td style="text-align:right;"> 0.6027 </td>
</tr>
</tbody>
</table>

```

```

# =====
# QUESTION 10: Improve the Worst Ensemble Model #####
# =====
cat("\n===== QUESTION 10: IMPROVE WORST ENSEMBLE =====\n")

```

```
===== QUESTION 10: IMPROVE WORST ENSEMBLE =====
```

```
cat("Worst ensemble identified:", worst_ens, "\n")
```

Worst ensemble identified: Bagging

```

# --- Attribute selection: union of top-15 RF features per class variable ---
top_attrs_list <- lapply(class_vars, function(cv) {
  rf_imp <- importance(get(paste0("rf_", cv)))[, "MeanDecreaseGini"]
  names(sort(rf_imp, decreasing = TRUE))[1:15]
})

```

```

''
top_attrs <- unique(unlist(top_attrs_list))
top_attrs <- intersect(top_attrs, pred_cols)
cat("Top attributes selected for improved model:", length(top_attrs), "\n")

```

Top attributes selected for improved model: 16

```
cat(paste(top_attrs, collapse = ", "), "\n")
```

Age, LifeSatis, GovtRespons, PrivateState, HardWork, Competition, PolLeader, ILPolitics, PolExperts, ILReligion, ILLeisure, PolArmy, ILFriends, FutureWork, ILWork, PolDemoc

```

improved_results <- list()
improved_probs <- list()

for (cv in class_vars) {
  cat("\n--- Improved model for:", cv, "---\n")
  fml_imp <- make_formula(cv, top_attrs)

  if (worst_ens == "Boosting") {
    imp.model <- adabag::boosting(fml_imp, data = WD.train,
                                mfinal = 100, coeflearn = "Breiman")
    imp.raw <- adabag::predict.boosting(imp.model, newdata = WD.test)
    imp.class <- factor(imp.raw$class, levels = c("Low", "High"))
    imp.prob <- imp.raw$prob[, 2] # col 2 = "High"

  } else if (worst_ens == "Bagging") {
    imp.model <- adabag::bagging(fml_imp, data = WD.train, mfinal = 100)
    imp.raw <- adabag::predict.bagging(imp.model, newdata = WD.test)
    imp.class <- factor(imp.raw$class, levels = c("Low", "High"))
    imp.prob <- imp.raw$prob[, 2] # col 2 = "High"

  } else {
    # Random Forest
    imp.model <- randomForest(fml_imp, data = WD.train,
                              ntree = 200,
                              mtry = floor(sqrt(length(top_attrs))),
                              importance = TRUE)
    imp.class <- predict(imp.model, WD.test)
    imp.prob <- predict(imp.model, WD.test, type = "prob")[, "High"]
  }

  actual <- WD.test[[cv]]
  cm_imp <- table(Predicted = imp.class, Actual = actual)
  cat("Confusion matrix (improved):\n")
  print(cm_imp)

  metrics_imp <- get_metrics(cm_imp)
  orig_f1 <- results_list[[cv]][worst_ens, "F1"]
  cat(sprintf(" Original %s F1: % Af | Improved F1: % Af\n",

```

```
cat(sprintf("Original %s F1: %.4f | Improved F1: %.4f\n",
           worst_ens, orig_f1, metrics_imp["F1"]))

improved_results[[cv]] <- metrics_imp
improved_probs[[cv]]  <- imp_prob
}
```

--- Improved model for: CWomensMvt ---

Confusion matrix (improved):

	Actual	
Predicted	Low	High
Low	115	112
High	1619	2388

Original Bagging F1: 0.7300 | Improved F1: 0.7340

--- Improved model for: CGovernment ---

Confusion matrix (improved):

	Actual	
Predicted	Low	High
Low	1624	1095
High	673	842

Original Bagging F1: 0.4329 | Improved F1: 0.4878

--- Improved model for: CUnions ---

Confusion matrix (improved):

	Actual	
Predicted	Low	High
Low	2330	1454
High	205	245

Original Bagging F1: 0.2231 | Improved F1: 0.2280

```
# ROC curves: original vs improved
cat("\n--- ROC curves: original vs improved ---\n")
```

--- ROC curves: original vs improved ---

```
for (cv in class_vars) {
  actual_bin <- ifelse(WD.test[[cv]] == "High", 1, 0)
  orig_prob  <- prob_list[[cv]][[worst_ens]]
  imp_prob   <- improved_probs[[cv]]

  prd_orig <- ROCR::prediction(orig_prob, actual_bin)
  prd_imp  <- ROCR::prediction(imp_prob,  actual_bin)
  prf_orig <- ROCR::performance(prd_orig, "tpr", "fpr")
  prf_imp  <- ROCR::performance(prd_imp,  "tpr", "fpr")
  auc_orig <- round(ROCR::performance(prd_orig, "auc")@y.values[[1]], 4)
  auc_imp  <- round(ROCR::performance(prd_imp,  "auc")@y.values[[1]], 4)
```

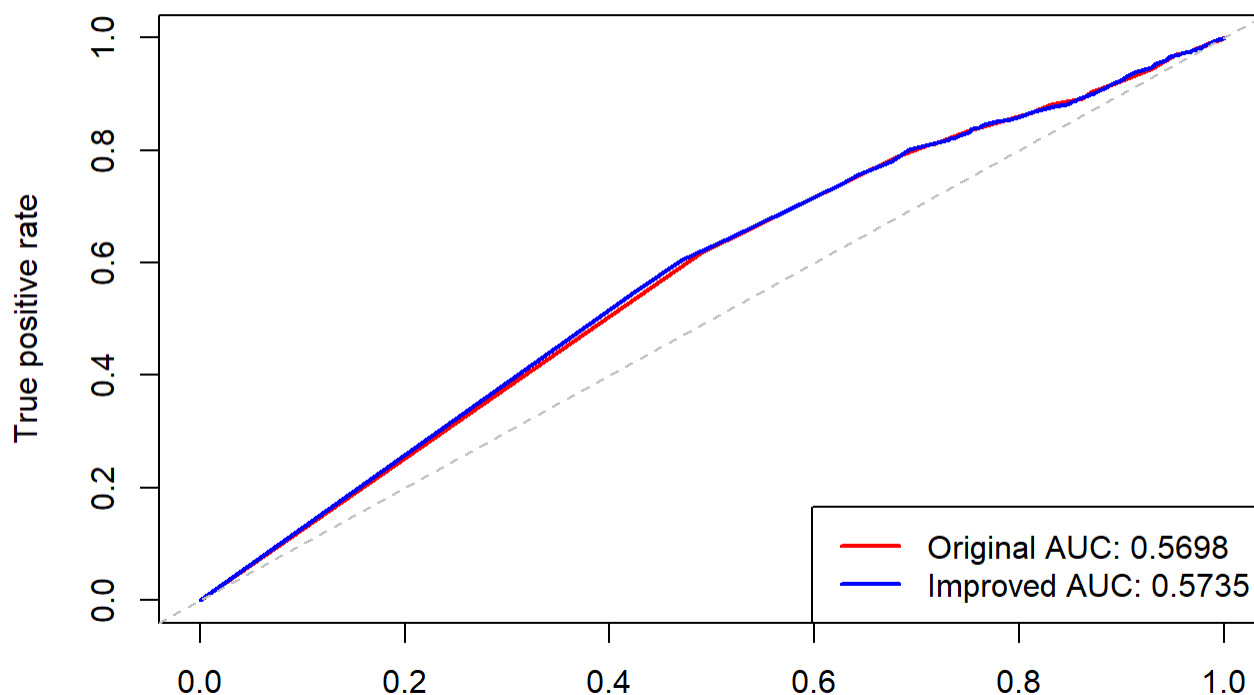
```

plot(prf_orig, col = "red", lwd = 2,
     main = paste("Figure: Q10 Improved", worst_ens, "-", cv))
plot(prf_imp, col = "blue", lwd = 2, add = TRUE)
abline(0, 1, lty = 2, col = "grey")
legend("bottomright",
      legend = c(paste("Original AUC:", auc_orig),
                  paste("Improved AUC:", auc_imp)),
      col = c("red", "blue"), lwd = 2)

# Comparison table
orig_m <- results_list[[cv]][worst_ens, ]
comp_df <- data.frame(
  Model      = c("Original", "Improved"),
  Accuracy   = c(orig_m["Accuracy"], improved_results[[cv]]["Accuracy"]),
  Precision  = c(orig_m["Precision"], improved_results[[cv]]["Precision"]),
  Recall     = c(orig_m["Recall"], improved_results[[cv]]["Recall"]),
  F1         = c(orig_m["F1"], improved_results[[cv]]["F1"]),
  AUC        = c(auc_orig, auc_imp)
)
print(kable(comp_df,
            caption = paste("Table: Q10 Original vs Improved -", cv),
            digits = 4) %>%
      kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                    full_width = FALSE))
}

```

Figure: Q10 Improved Bagging – CWomensMvt



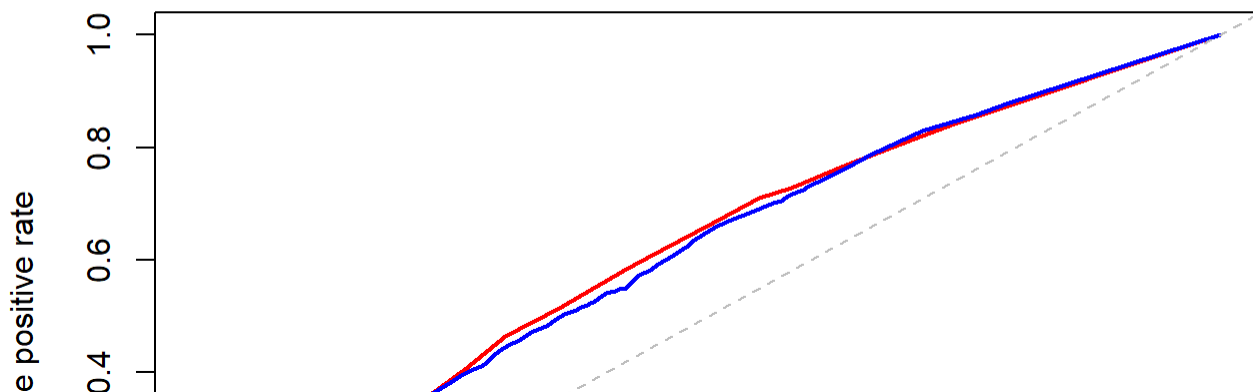
False positive rate

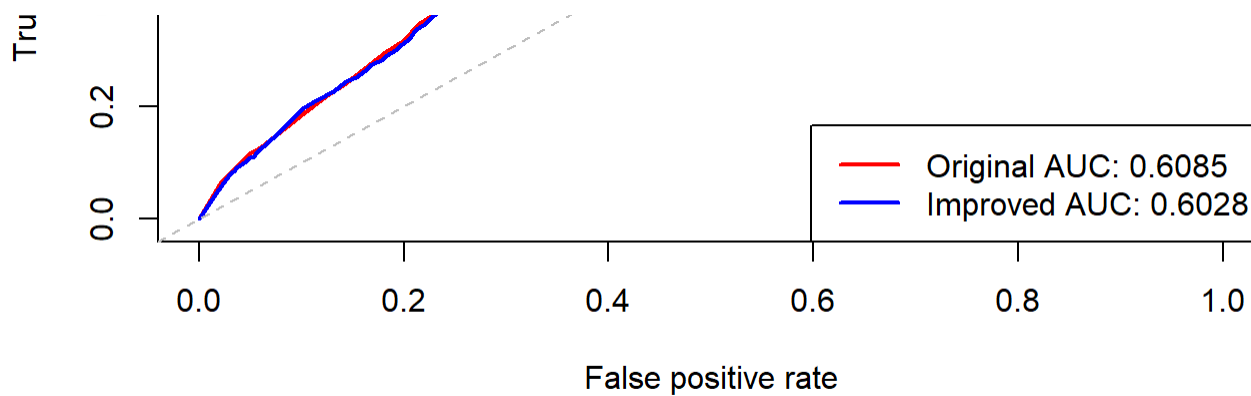
```

<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">
<caption>Table: Q10 Original vs Improved - CWomensMvt</caption>
<thead>
<tr>
<th style="text-align:left;"> Model </th>
<th style="text-align:right;"> Accuracy </th>
<th style="text-align:right;"> Precision </th>
<th style="text-align:right;"> Recall </th>
<th style="text-align:right;"> F1 </th>
<th style="text-align:right;"> AUC </th>
</tr>
</thead>
<tbody>
<tr>
<td style="text-align:left;"> Original </td>
<td style="text-align:right;"> 0.5869 </td>
<td style="text-align:right;"> 0.5944 </td>
<td style="text-align:right;"> 0.9456 </td>
<td style="text-align:right;"> 0.730 </td>
<td style="text-align:right;"> 0.5698 </td>
</tr>
<tr>
<td style="text-align:left;"> Improved </td>
<td style="text-align:right;"> 0.5912 </td>
<td style="text-align:right;"> 0.5960 </td>
<td style="text-align:right;"> 0.9552 </td>
<td style="text-align:right;"> 0.734 </td>
<td style="text-align:right;"> 0.5735 </td>
</tr>
</tbody>
</table>

```

Figure: Q10 Improved Bagging – CGovernment





```
<table class="table table-striped table-hover table-condensed" style="width: auto !important; margin-left: auto; margin-right: auto;">
```

```
<caption>Table: Q10 Original vs Improved - CGovernment</caption>
```

```
<thead>
```

```
<tr>
```

```
<th style="text-align:left;"> Model </th>
```

```
<th style="text-align:right;"> Accuracy </th>
```

```
<th style="text-align:right;"> Precision </th>
```

```
<th style="text-align:right;"> Recall </th>
```

```
<th style="text-align:right;"> F1 </th>
```

```
<th style="text-align:right;"> AUC </th>
```

```
</tr>
```

```
</thead>
```

```
<tbody>
```

```
<tr>
```

```
<td style="text-align:left;"> Original </td>
```

```
<td style="text-align:right;"> 0.5848 </td>
```

```
<td style="text-align:right;"> 0.5770 </td>
```

```
<td style="text-align:right;"> 0.3464 </td>
```

```
<td style="text-align:right;"> 0.4329 </td>
```

```
<td style="text-align:right;"> 0.6085 </td>
```

```
</tr>
```

```
<tr>
```

```
<td style="text-align:left;"> Improved </td>
```

```
<td style="text-align:right;"> 0.5824 </td>
```

```
<td style="text-align:right;"> 0.5558 </td>
```

```
<td style="text-align:right;"> 0.4347 </td>
```

```
<td style="text-align:right;"> 0.4878 </td>
```

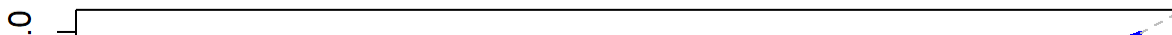
```
<td style="text-align:right;"> 0.6028 </td>
```

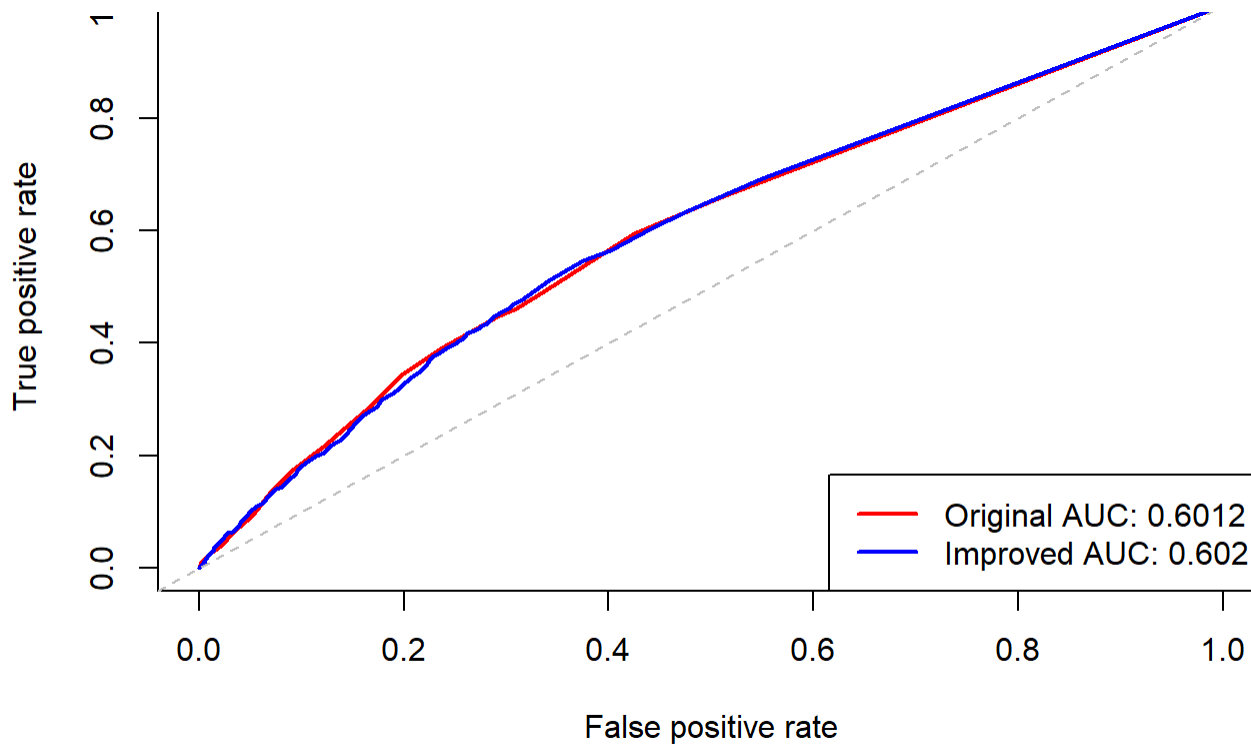
```
</tr>
```

```
</tbody>
```

```
</table>
```

Figure: Q10 Improved Bagging – CUnions





```
<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">
```

```
<caption>Table: Q10 Original vs Improved - CUnions</caption>
```

```
<thead>
```

```
<tr>
```

```
<th style="text-align:left;"> Model </th>
```

```
<th style="text-align:right;"> Accuracy </th>
```

```
<th style="text-align:right;"> Precision </th>
```

```
<th style="text-align:right;"> Recall </th>
```

```
<th style="text-align:right;"> F1 </th>
```

```
<th style="text-align:right;"> AUC </th>
```

```
</tr>
```

```
</thead>
```

```
<tbody>
```

```
<tr>
```

```
<td style="text-align:left;"> Original </td>
```

```
<td style="text-align:right;"> 0.6117 </td>
```

```
<td style="text-align:right;"> 0.5659 </td>
```

```
<td style="text-align:right;"> 0.1389 </td>
```

```
<td style="text-align:right;"> 0.2231 </td>
```

```
<td style="text-align:right;"> 0.6012 </td>
```

```
</tr>
```

```
<tr>
```

```
<td style="text-align:left;"> Improved </td>
```

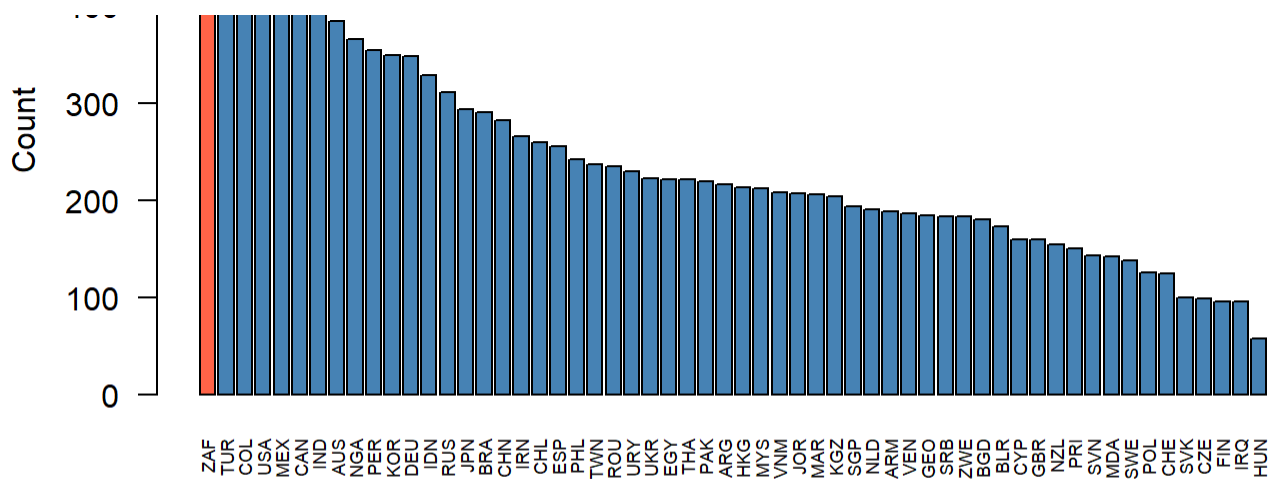
```
<td style="text-align:right;"> 0.6082 </td>
```

```
<td style="text-align:right;"> 0.5444 </td>
```

```
<td style="text-align:right;"> 0.1442 </td>
```

```
<td style="text-align:right;"> 0.2280 </td>
```

```
<td style="text-align:right;"> 0.6020 </td>
```

```
par(mar = c(5, 4, 4, 2))

# Select class variable for ANN
# Using first class variable (CWomensMvt) based on Q8 insights
cv_ann <- class_vars[1]
cat("Class variable for ANN:", cv_ann, "\n")
```

Class variable for ANN: CWomensMvt

```
# Step 3: Subset ALL cleaned data for selected country
WD_country_all <- WD_clean[WD_clean$Country == top_country, ]
cat("Total observations for", top_country, ":", nrow(WD_country_all), "\n")
```

Total observations for ZAF : 548

```
# Also prepare test subset for Q12 wave evaluation
WD_test_country <- WD.test[WD.test$Country == top_country, ]
cat("Test observations for", top_country, ":", nrow(WD_test_country), "\n")
```

Test observations for ZAF : 163

```
# Select top predictors from Q8 feature importance
top_ann_attrs <- unique(unlist(lapply(class_vars, function(cv) {
  rf_imp <- importance(get(paste0("rf_", cv)))[, "MeanDecreaseGini"]
  names(sort(rf_imp, decreasing = TRUE))[1:10]
})))
top_ann_attrs <- intersect(top_ann_attrs, pred_cols)
cat("Number of selected predictors for ANN:", length(top_ann_attrs), "\n")
```

Number of selected predictors for ANN: 11

```
cat("Selected predictors:", paste(top_ann_attrs, collapse = ", "), "\n")
```

Selected predictors: Age, LifeSatis, GovtRespons, PrivateState, HardWork, Competition, PolLeader, ILPolitics, PolExperts, ILReligion, PolArmy

```
# ANN Pre-processing – standardise numeric predictors
num_ann_cols <- top_ann_attrs[sapply(WD_country_all[top_ann_attrs], is.numeric)]

sc_mean      <- apply(WD_country_all[num_ann_cols], 2, mean)
sc_sd        <- apply(WD_country_all[num_ann_cols], 2, sd)
sc_sd[sc_sd == 0] <- 1 # prevent division by zero for constant columns

# Scale ALL country data (used for training in Q11)
WD_ann_all <- WD_country_all
WD_ann_all[num_ann_cols] <- scale(WD_country_all[num_ann_cols],
                                center = sc_mean,
                                scale = sc_sd)

# Add WaveNum: numeric version of Wave for temporal modelling (Q12)
WD_ann_all$WaveNum <- as.numeric(as.character(WD_ann_all$Wave))

# Scale test country data using SAME parameters (no re-fitting)
WD_test_ann <- WD_test_country
WD_test_ann[num_ann_cols] <- scale(WD_test_country[num_ann_cols],
                                center = sc_mean,
                                scale = sc_sd)
WD_test_ann$WaveNum <- as.numeric(as.character(WD_test_ann$Wave))

# Final predictor set (scaled numeric + WaveNum)
ann_preds <- c(num_ann_cols, "WaveNum")
cat("Final ANN predictor count:", length(ann_preds), "\n")
```

Final ANN predictor count: 12

```
# Build and train ANN on ALL country data
# Architecture: 1 hidden layer, 5 hidden units
# - nnet uses logistic (sigmoid) activation by default
# - decay = 0.01 is L2 regularisation to reduce overfitting
# - maxit = 300 allows sufficient convergence
# - size = 5 chosen as a balance between capacity and overfitting
#   given the moderate dataset size for one country
fml_ann <- make_formula(cv_ann, ann_preds)

set.seed(33986010)
ann.model <- nnet(fml_ann,
```

```

data = WD_ann_all, # ALL country data, not just 10%
size = 5,
maxit = 300,
decay = 0.01,
trace = FALSE)

```

```

cat("ANN architecture: input =", length(ann_preds),
    "| hidden = 5 | output = 1 (sigmoid)\n")

```

ANN architecture: input = 12 | hidden = 5 | output = 1 (sigmoid)

```

cat("Total weights:", length(ann.model$wts), "\n")

```

Total weights: 71

```

# Evaluate on full country test set
ann.prob <- predict(ann.model, WD_test_ann)
ann.class <- factor(ifelse(ann.prob > 0.5, "High", "Low"),
                    levels = c("Low", "High"))

actual_ann <- WD_test_ann[[cv_ann]]
cm_ann <- table(Predicted = ann.class, Actual = actual_ann)
cat("\nANN Confusion Matrix (", top_country, "):\n")

```

ANN Confusion Matrix (ZAF):

```

print(cm_ann)

```

	Actual	
Predicted	Low	High
Low	42	11
High	25	85

```

metrics_ann <- get_metrics(cm_ann)

print(
  kable(data.frame(Metric = names(metrics_ann),
                  Value = as.numeric(metrics_ann)),
        caption = paste("Table: Q11 ANN Metrics -", top_country, "|", cv_ann),
        digits = 4) %>%
    kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
                  full_width = FALSE) %>%
    row_spec(0, bold = TRUE, background = "#4472C4", color = "white")
)

```

```

<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">

```

```

<caption>Table: Q11 ANN Metrics - ZAF | CWomensMvt</caption>
<thead>
  <tr>
    <th style="text-align:left;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Metric </th>
    <th style="text-align:right;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Value </th>
  </tr>
</thead>
<tbody>
  <tr>
    <td style="text-align:left;"> Accuracy </td>
    <td style="text-align:right;"> 0.7791 </td>
  </tr>
  <tr>
    <td style="text-align:left;"> Precision </td>
    <td style="text-align:right;"> 0.7727 </td>
  </tr>
  <tr>
    <td style="text-align:left;"> Recall </td>
    <td style="text-align:right;"> 0.8854 </td>
  </tr>
  <tr>
    <td style="text-align:left;"> F1 </td>
    <td style="text-align:right;"> 0.8252 </td>
  </tr>
</tbody>
</table>

```

```

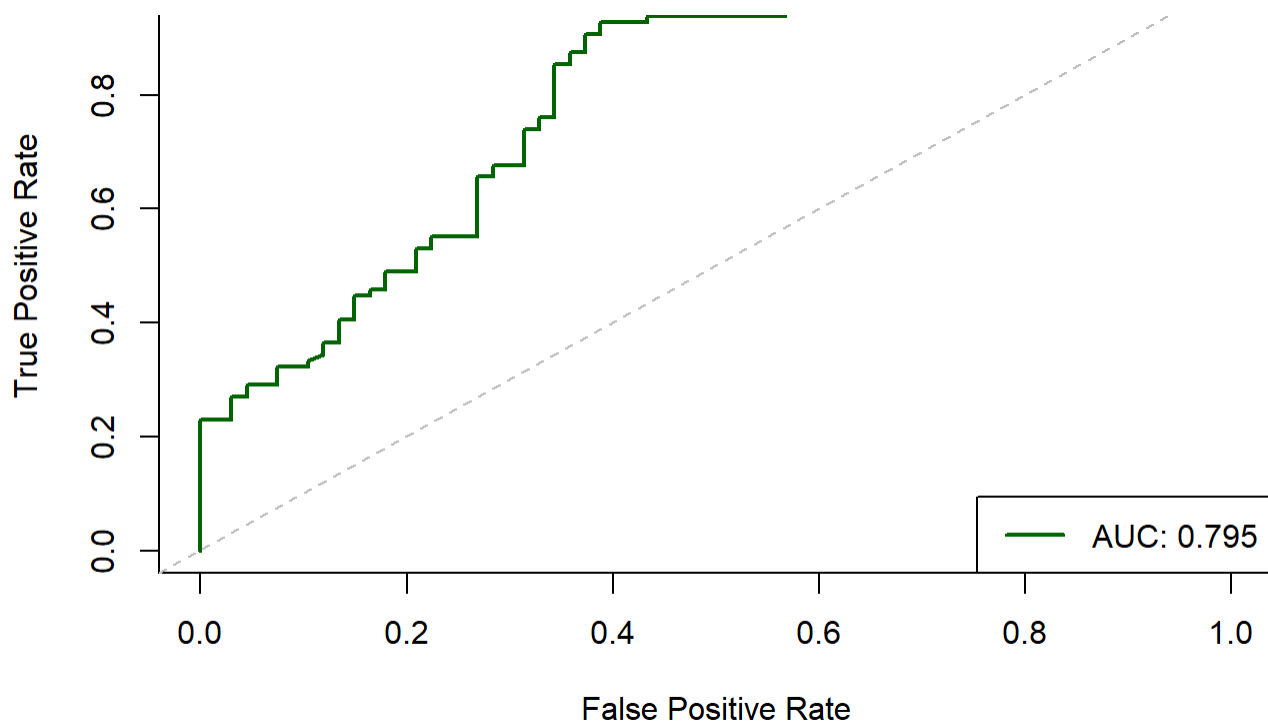
# ROC Curve for ANN
actual_bin_ann <- ifelse(actual_ann == "High", 1, 0)
prd_ann        <- ROCR::prediction(ann.prob, actual_bin_ann)
prf_ann        <- ROCR::performance(prd_ann, "tpr", "fpr")
auc_ann        <- round(ROCR::performance(prd_ann, "auc")@y.values[[1]], 4)

plot(prf_ann, col = "darkgreen", lwd = 2,
     main = paste("Figure: Q11 ANN ROC -", top_country, "|", cv_ann),
     xlab = "False Positive Rate",
     ylab = "True Positive Rate")
abline(0, 1, lty = 2, col = "grey")
legend("bottomright",
      legend = paste("AUC:", auc_ann),
      col    = "darkgreen", lwd = 2)

```

Figure: Q11 ANN ROC – ZAF | CWomensMvt





```
cat("Q11 ANN AUC:", auc_ann, "\n")
```

Q11 ANN AUC: 0.795

```
# =====
# QUESTION 12: ANN Performance Over Two Waves
# =====
cat("\n===== QUESTION 12: ANN PERFORMANCE BY WAVE =====\n")
```

===== QUESTION 12: ANN PERFORMANCE BY WAVE =====

```
# Identify the two waves with most TEST observations
# for the selected country

wave_counts_test <- sort(table(WD_test_ann$Wave), decreasing = TRUE)
wave_counts_test <- wave_counts_test[wave_counts_test > 0] # drop empty levels

cat("Test observations per wave for", top_country, ":\n")
```

Test observations per wave for ZAF :

```
print(wave_counts_test)
```

```
5 6 4 3
-- -- -- --
```

49 43 36 35

```
top2_waves <- names(head(wave_counts_test, 2))
cat("Top 2 waves selected:", paste(top2_waves, collapse = ", "), "\n")
```

Top 2 waves selected: 5, 6

```
# Evaluate ANN separately for each of the two waves
# The SAME trained model from Q11 is used (no retraining)
wave_metrics_list <- list()
wave_auc_list      <- list()
wave_prob_list     <- list()
wave_actual_list   <- list()

col_wave <- c("blue", "red")

par(mfrow = c(1, 2))

for (i in seq_along(top2_waves)) {
  wv <- top2_waves[i]
  tdat <- WD_test_ann[WD_test_ann$Wave == wv, ]
  cat("\n--- Wave", wv, "| N test:", nrow(tdat), "---\n")

  # Predict using Q11 model
  wp <- predict(ann.model, tdat)
  wc <- factor(ifelse(wp > 0.5, "High", "Low"), levels = c("Low", "High"))
  wa <- tdat[[cv_ann]]

  # Confusion matrix
  cm_w <- table(Predicted = wc, Actual = wa)
  cat("Confusion Matrix - Wave", wv, ":\n")
  print(cm_w)

  # Metrics
  m_w <- get_metrics(cm_w)
  wave_metrics_list[[wv]] <- m_w

  # ROC & AUC
  actual_bin_w <- ifelse(wa == "High", 1, 0)
  prd_w <- ROCR::prediction(wp, actual_bin_w)
  prf_w <- ROCR::performance(prd_w, "tpr", "fpr")
  auc_w <- round(ROCR::performance(prd_w, "auc")@y.values[[1]], 4)
  wave_auc_list[[wv]] <- auc_w
  wave_prob_list[[wv]] <- wp
  wave_actual_list[[wv]] <- actual_bin_w

  # Individual ROC plot (side by side)
  plot(prf_w, col = col_wave[i], lwd = 2,
       main = paste("Figure: Q12 Wave", wv, "-", top_country),
       plot = "F1 - Precision - Recall")
}
```

```

xlab = "False Positive Rate",
ylab = "True Positive Rate")
abline(0, 1, lty = 2, col = "grey")
legend("bottomright",
      legend = paste("AUC:", auc_w),
      col     = col_wave[i], lwd = 2)

cat("Wave", wv, "AUC:", auc_w, "\n")
}

```

--- Wave 5 | N test: 49 ---

Confusion Matrix - Wave 5 :

	Actual	
Predicted	Low	High
Low	10	3
High	11	25

Wave 5 AUC: 0.7347

--- Wave 6 | N test: 43 ---

Confusion Matrix - Wave 6 :

	Actual	
Predicted	Low	High
Low	13	4
High	5	21

Figure: Q12 Wave 5 – ZAF

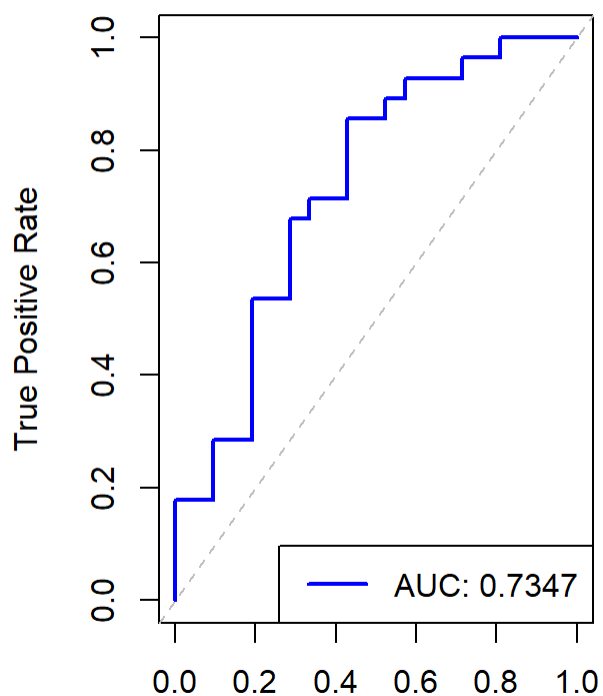
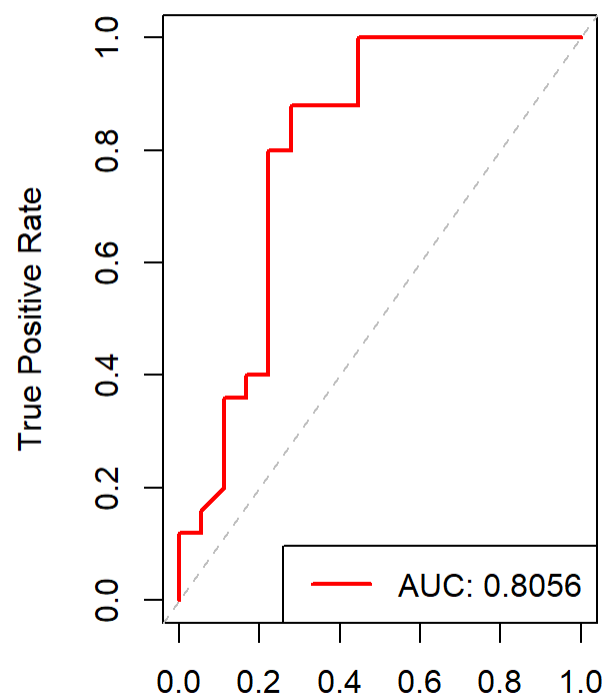


Figure: Q12 Wave 6 – ZAF



False Positive Rate

False Positive Rate

Wave 6 AUC: 0.8056

```

par(mfrow = c(1, 1))

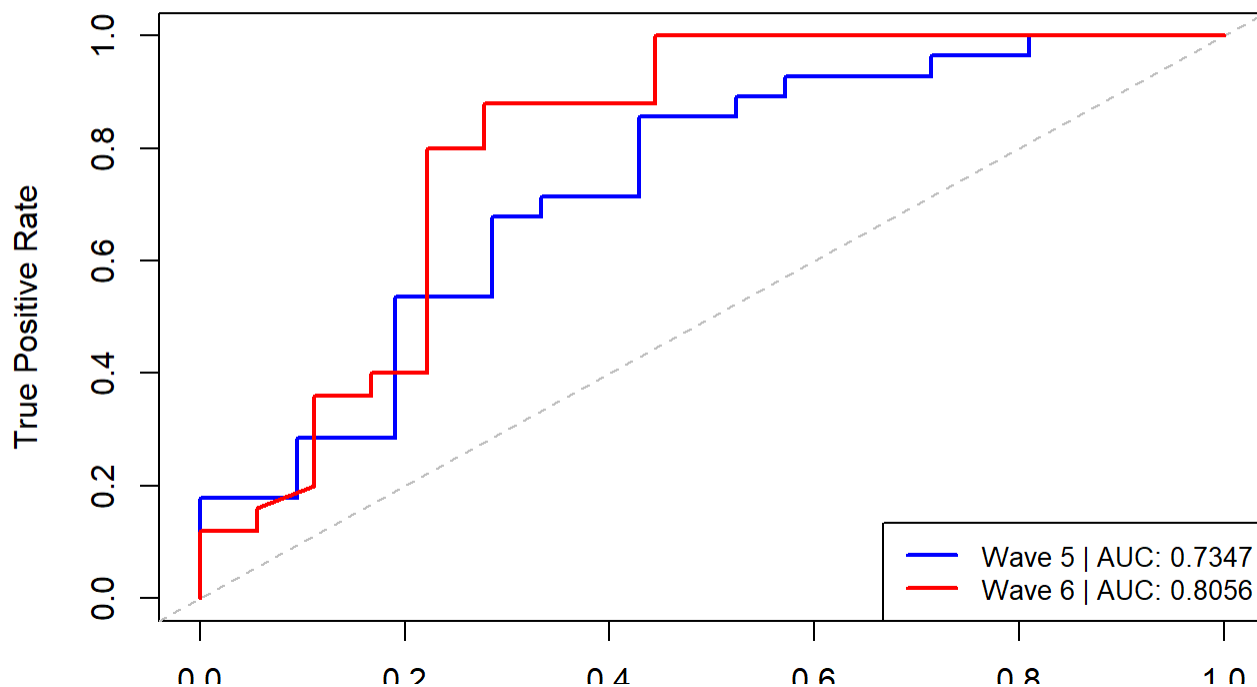
# Combined ROC plot – both waves on same axis
plot(0, 0, type = "n", xlim = c(0,1), ylim = c(0,1),
     xlab = "False Positive Rate",
     ylab = "True Positive Rate",
     main = paste("Figure: Q12 ANN ROC by Wave -", top_country, "|", cv_ann))
abline(0, 1, lty = 2, col = "grey")

for (i in seq_along(top2_waves)) {
  wv <- top2_waves[i]
  prd_w <- ROCR::prediction(wave_prob_list[[wv]], wave_actual_list[[wv]])
  prf_w <- ROCR::performance(prd_w, "tpr", "fpr")
  plot(prf_w, col = col_wave[i], lwd = 2, add = TRUE)
}

legend("bottomright",
      legend = paste("Wave", top2_waves,
                    "| AUC:", sapply(top2_waves,
                                     function(w) wave_auc_list[[w]])),
      col = col_wave, lwd = 2, cex = 0.85)

```

Figure: Q12 ANN ROC by Wave – ZAF | CWomensMvt



0.0

0.2

0.4

0.6

0.8

1.0

False Positive Rate

```
# Summary comparison table across the two waves
wave_summary_df <- do.call(rbind, lapply(top2_waves, function(wv) {
  data.frame(
    Wave      = wv,
    N_Test    = sum(WD_test_ann$Wave == wv),
    Accuracy   = wave_metrics_list[[wv]]["Accuracy"],
    Precision  = wave_metrics_list[[wv]]["Precision"],
    Recall    = wave_metrics_list[[wv]]["Recall"],
    F1        = wave_metrics_list[[wv]]["F1"],
    AUC       = wave_auc_list[[wv]]
  )
}))
rownames(wave_summary_df) <- NULL

cat("\nQ12: ANN Performance Comparison by Wave\n")
```

Q12: ANN Performance Comparison by Wave

```
print(
  kable(wave_summary_df,
    caption = paste("Table: Q12 ANN Performance by Wave -",
      top_country, "|", cv_ann),
    digits = 4,
    align = c("c","c","c","c","c","c","c")) %>%
  kable_styling(bootstrap_options = c("striped", "hover", "condensed"),
    full_width = FALSE) %>%
  row_spec(0, bold = TRUE, background = "#4472C4", color = "white") %>%
  column_spec(1, bold = TRUE)
)
```

```
<table class="table table-striped table-hover table-condensed" style="width: auto !important;
margin-left: auto; margin-right: auto;">
```

```
<caption>Table: Q12 ANN Performance by Wave - ZAF | CWomensMvt</caption>
```

```
<thead>
```

```
<tr>
```

```
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Wave </th>
```

```
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> N_Test </th>
```

```
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
rgba(68, 114, 196, 255) !important;"> Accuracy </th>
```

```
<th style="text-align:center;font-weight: bold;color: white !important;background-color:
```

```

    rgba(68, 114, 196, 255) !important;"} Precision </th>
    <th style="text-align:center;font-weight: bold;color: white !important;background-color:
    rgba(68, 114, 196, 255) !important;"> Recall </th>
    <th style="text-align:center;font-weight: bold;color: white !important;background-color:
    rgba(68, 114, 196, 255) !important;"> F1 </th>
    <th style="text-align:center;font-weight: bold;color: white !important;background-color:
    rgba(68, 114, 196, 255) !important;"> AUC </th>
  </tr>
</thead>
<tbody>
  <tr>
    <td style="text-align:center;font-weight: bold;"> 5 </td>
    <td style="text-align:center;"> 49 </td>
    <td style="text-align:center;"> 0.7143 </td>
    <td style="text-align:center;"> 0.6944 </td>
    <td style="text-align:center;"> 0.8929 </td>
    <td style="text-align:center;"> 0.7812 </td>
    <td style="text-align:center;"> 0.7347 </td>
  </tr>
  <tr>
    <td style="text-align:center;font-weight: bold;"> 6 </td>
    <td style="text-align:center;"> 43 </td>
    <td style="text-align:center;"> 0.7907 </td>
    <td style="text-align:center;"> 0.8077 </td>
    <td style="text-align:center;"> 0.8400 </td>
    <td style="text-align:center;"> 0.8235 </td>
    <td style="text-align:center;"> 0.8056 </td>
  </tr>
</tbody>
</table>

```

```

# Change summary (difference between waves)
if (length(top2_waves) == 2) {
  delta_f1 <- round(wave_metrics_list[[top2_waves[2]]]["F1"] -
                    wave_metrics_list[[top2_waves[1]]]["F1"], 4)
  delta_auc <- round(wave_auc_list[[top2_waves[2]]] -
                     wave_auc_list[[top2_waves[1]]], 4)
  cat(sprintf("\nChange from Wave %s to Wave %s:\n",
              top2_waves[1], top2_waves[2]))
  cat(sprintf("  Delta F1   : %+.4f\n", delta_f1))
  cat(sprintf("  Delta AUC  : %+.4f\n", delta_auc))
}

```

Change from Wave 5 to Wave 6:

```

Delta F1   : +0.0423
Delta AUC  : +0.0709

```